# McDonnell Douglas F-4 Phantom II

**Contents** hide

(Top)
Development
Design
Operational history
Culture
Variants
Operators
Notable accidents
Aircraft on display
Gallery
Specifications (F-4E)
See also
References
External links

**Appearance** hide

Text
○ Small
● Standard
○ Large

Width
● Standard
○ Wide

Color
○ Automatic
● Light
○ Dark

From Wikipedia, the free encyclopedia

*"F-4" redirects here. For other uses, see F4.*

The **McDonnell Douglas F-4 Phantom II**[N 1] is an American tandem two-seat, twin-engine, all-weather, long-range supersonic jet interceptor and fighter-bomber that was developed by McDonnell Aircraft for the United States Navy.[4] It entered service with the Navy in 1961,[5] then was adopted by the United States Marine Corps, and the United States Air Force, and within a few years became a major part of their air arms.[6] A total of 5,195 Phantoms were built from 1958 to 1981, making it the most-produced American supersonic military aircraft in history and a signature combat aircraft of the Cold War.[6][7]

The Phantom is a large fighter with a top speed of over Mach 2.2. It can carry more than 18,000 pounds (8,200 kg) of weapons on nine external hardpoints, including air-to-air missiles, air-to-ground missiles, and various bombs.[8] Like some other interceptors of its time, the F-4 was initially designed without an internal cannon, but some later models incorporated an internal M61 Vulcan rotary cannon. Beginning in 1959, it set 15 world records for in-flight performance,[9] including an absolute speed record and an absolute altitude record.[10]

The F-4 was used extensively during the Vietnam War, first as the principal air superiority fighter for the U.S. Air Force, Navy, and Marine Corps, and later as a ground-attack and aerial reconnaissance aircraft. During the Vietnam War, all five American servicemen who became aces – one U.S. Air Force pilot and two weapon systems officers (WSOs),[11] one U.S. Navy pilot and one radar intercept officer (RIO) – did so in F-4s.[12] The Phantom remained a major part of U.S. military air power into the 1980s, when it was gradually replaced by more modern aircraft such as the F-15 Eagle and F-16 Fighting Falcon in the U.S. Air Force, the F-14 Tomcat in the U.S. Navy, and the F/A-18 Hornet in the U.S. Navy and U.S. Marine Corps.

The F-4G Phantom was used for reconnaissance and Wild Weasel (Suppression of Enemy Air Defenses) missions in the 1991 Gulf War, and finally left combat service in 1996.[13][14] It was the only aircraft used by both U.S. flight demonstration teams: the United States Air Force Thunderbirds (F-4E) and the United States Navy Blue Angels (F-4J).[6][15][16] The F-4 was also operated by the armed forces of 11 other nations. Israeli Phantoms saw extensive combat in several Arab–Israeli conflicts, while Iran used its large fleet of Phantoms, acquired before the fall of the Shah, in the Iran–Iraq War. The F-4 remains in active service with the Hellenic Air Force[17] and the Turkish Air Force. Turkey's most recently upgraded F-4E Terminator variant is to remain in service until at least 2030.[18]

**F-4 Phantom II**

A U.S. Air Force QF-4E flies with the 82nd Aerial Targets Squadron over White Sands Missile Range in 2008

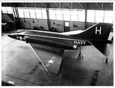
| General information | |
|---|---|
| **Type** | Interceptor, fighter-bomber |
| **National origin** | United States |
| **Manufacturer** | McDonnell Aircraft Corporation<br>McDonnell Douglas |
| **Status** | In limited service |
| **Primary users** | United States Air Force (historical)<br>United States Navy (historical)<br>United States Marine Corps (historical)<br>Hellenic Air Force |
| **Number built** | 5,195 |
| **History** | |
| **Manufactured** | 1958–1981 |
| **Introduction date** | 30 December 1960; 65 years ago |
| **First flight** | 27 May 1958; 68 years ago |
| **Retired** | 1986 (USN)<br>1996 (U.S. combat use)<br>2004 (Israel)<br>2013 (Germany)<br>2016 (U.S. target drone)[1]<br>2021 (Japan)[2]<br>2024 (South Korea)[3] |
| **Variant** | McDonnell Douglas Phantom FG.1/FGR.2 |

## Development [edit]

### Origins [edit]

In 1952, McDonnell's Chief of Aerodynamics, Dave Lewis, was appointed by CEO Jim McDonnell to be the company's preliminary design manager.[19] With no new aircraft competitions on the horizon, internal studies concluded the Navy had the greatest need for a new and different aircraft type: an attack fighter.[20]

In 1953, McDonnell Aircraft began work on revising its F3H Demon naval fighter, seeking expanded capabilities and better performance. The company developed several projects, including a variant powered by a Wright J67 engine,[21] and variants powered by two Wright J65 engines, or two General Electric J79 engines.[22] The J79-powered version promised a top speed of Mach 1.97. On 19 September 1953, McDonnell approached the United States Navy with a proposal for the "Super Demon". Uniquely, the aircraft was to be modular, as it could be fitted with one- or two-seat noses for different missions, with different nose cones to accommodate radar, photo cameras, four 20 mm (.79 in) cannon, or 56 FFAR unguided rockets in addition to the nine hardpoints under the wings and the fuselage. The Navy was sufficiently interested to order a full-scale mock-up of the F3H-G/H, but felt that the upcoming Grumman XF9F-9 and Vought XF8U-1 already satisfied the need for a supersonic fighter.[23]



The McDonnell F3H-G/H mockup, 1954

The McDonnell design was therefore reworked into an all-weather fighter-bomber with 11 external hardpoints for weapons and on 18 October 1954, the company received a letter of intent for two YAH-1 prototypes. Then on 26 May 1955, four Navy officers arrived at the McDonnell offices and, within an hour, presented the company with an entirely new set of requirements. Because the Navy already had the Douglas A-4 Skyhawk for ground attack and F-8 Crusader for dogfighting, the project now had to fulfill the need for an all-weather fleet defense interceptor. A second crewman was added to operate the powerful radar;[4] designers believed that air combat in the next war would overload solo pilots with information.[24]

### XF4H-1 prototype [edit]

Key figures in the F-4 development: David Lewis, Robert Little, and Herman Barkey

The XF4H-1 was designed to carry four semi-recessed AAM-N-6 Sparrow III radar-guided missiles, and to be powered by two J79-GE-8 engines. As in the McDonnell F-101 Voodoo, the engines sat low in the fuselage to maximize internal fuel capacity and ingested air through fixed geometry intakes. The thin-section wing had a leading edge sweep of 45° and was equipped with blown flaps for better low-speed handling.[25]

Wind tunnel testing had revealed lateral instability, requiring the addition of 5° dihedral to the wings.[26] To avoid redesigning the titanium central section of the aircraft, McDonnell engineers angled up only the outer portions of the wings by 12°, which averaged to the required 5° over the entire wingspan. The wings also received the distinctive "dogtooth" for improved control at high angles of attack. The all-moving tailplane was given 23° of anhedral to improve control at high angles of attack, while still keeping the tailplane clear of the engine exhaust.[25] In addition, air intakes were equipped with one fixed ramp and one variable geometry ramp with angle scheduled to give maximum pressure recovery between Mach 1.4 and Mach 2.2. Airflow matching between the inlet and engine was achieved by bypassing the engine as secondary air into the exhaust nozzle. All-weather intercept capability was achieved with the AN/APQ-50 radar. To meet requirements for carrier operations, the landing gear was designed to withstand landings with a maximum sink rate of 23 ft/s (7 m/s), while the nose strut could extend by 20 in (51 cm) to increase angle of attack on the catapult portion of a takeoff.[26]

On 25 July 1955, the Navy ordered two XF4H-1 test aircraft and five YF4H-1 pre-production examples. The Phantom made its maiden flight on 27 May 1958 with Robert C. Little at the controls. A hydraulic problem precluded the retraction of the landing gear, but subsequent flights went more smoothly. Early testing resulted in redesign of the air intakes, including the distinctive addition of 12,500 holes to "bleed off" the slow-moving boundary layer air from the surface of each intake ramp. Series production aircraft also featured splitter plates to divert the boundary layer away from the engine intakes. The aircraft was soon in competition with the XF8U-3 Crusader III. Due to cockpit workload, the Navy wanted a two-seat aircraft and on 17 December 1958 the F4H was declared the winner. Delays with the J79-GE-8 engines meant that the first production aircraft were fitted with J79-GE-2 and −2A engines, each having 16,100 lbf (71.8 kN) of afterburning thrust. In 1959, the Phantom began carrier suitability trials with the first complete launch-recovery cycle performed on 15 February 1960 from *Independence*.[26]

*The F4H Navy Fighter* (1960)
Official McDonnell promotional film reel

There were proposals to name the F4H "Satan" and "Mithras".[26] In the end, the aircraft was given the less controversial name "Phantom II", the first "Phantom" being another McDonnell jet fighter, the FH-1 Phantom. The Phantom II was briefly given the designation F-110A and named "Spectre" by the USAF[27] and the Tri-Service aircraft designation system, F-4, was adopted in September 1962.

An F4H-1F aboard *Independence*, April 1960

## Production

Early in production, the radar was upgraded to the Westinghouse AN/APQ-72, an AN/APQ-50 with a larger radar antenna, necessitating the bulbous nose, and the canopy was reworked to improve visibility and make the rear cockpit less claustrophobic.[28] During its career the Phantom underwent many changes in the form of numerous variants developed.

In 1961, VF-74 became the first operational U.S. Navy *Phantom* squadron

The USN operated the F4H-1 (re-designated F-4A in 1962) with J79-GE-2 and -2A engines of 16,100 lbf (71.62 kN) thrust and later builds receiving -8 engines. A total of 45 F-4As were built; none saw combat, and most ended up as test or training aircraft.[29] The USN and USMC received the first definitive Phantom, the F-4B which was equipped with the Westinghouse APQ-72 radar (pulse only), a Texas Instruments AN/AAA-4 Infrared search and track pod under the nose, an AN/AJB-3 bombing system and powered by J79-GE-8,-8A and -8B engines of 10,900 lbf (48.5 kN) dry and 16,950 lbf (75.4 kN) afterburner (reheat) with the first flight on 25 March 1961. 649 F-4Bs were built with deliveries beginning in 1961 and VF-121 *Pacemakers* receiving the first examples at NAS Miramar.[29]

The USAF received Phantoms as the result of Defense Secretary Robert McNamara's push to create a unified fighter for all branches of the US military. After an F-4B won the "Operation Highspeed" fly-off against the Convair F-106 Delta Dart, the USAF borrowed two Naval F-4Bs, temporarily designating them F-110A in January 1962, and developed requirements for their own version. Unlike the US Navy's focus on air-to-air interception in the Fleet Air Defense (FAD) mission, the USAF emphasized both an air-to-air and an air-to-ground fighter-bomber role. With McNamara's unification of designations on 18 September 1962, the Phantom became the F-4 with the naval version designated F-4B and USAF F-4C. The first Air Force Phantom flew on 27 May 1963, exceeding Mach 2 on its maiden flight.[30]

The F-4J improved both air-to-air and ground-attack capability; deliveries begun in 1966 and ended in 1972 with 522 built.[31] It was equipped with J79-GE-10 engines with 17,844 lbf (79.374 kN) thrust, the Westinghouse AN/AWG-10 Fire Control System (making the F-4J the first fighter in the world with operational look-down/shoot-down capability),[32] a new integrated missile control system and the AN/AJB-7 bombing system for expanded ground attack capability.[33]

The F-4N (updated F-4Bs) with smokeless engines and F-4J aerodynamic improvements started in 1972 under a U.S. Navy-initiated refurbishment program called "Project Bee Line"[34] with 228 converted by 1978. The F-4S model resulted from the refurbishment of 265 F-4Js with J79-GE-17 smokeless engines of 17,900 lbf (79.379 kN), AWG-10B radar with digitized circuitry for improved performance and reliability, Honeywell AN/AVG-8 Visual Target Acquisition Set or VTAS (world's first operational Helmet Sighting System), classified avionics improvements, airframe reinforcement and leading edge slats for enhanced maneuvering.[35] The USMC also operated the RF-4B with reconnaissance cameras with 46 built;[36] the RF-4B flew alone and unarmed, with a requirement to fly straight and level at 5,000 feet while taking photographs. They relied on the shortcomings of the anti-aircraft defenses to survive as they were unable to make evasive maneuvers.[24]

1960 US Navy F-4 Phantom promotional film, with foreword by test-pilot Bob Little.

Phantom II production ended in the United States in 1979 after 5,195 had been built (5,057 by McDonnell Douglas and 138 in Japan by Mitsubishi). Of these, 2,874 went to the USAF, 1,264 to the Navy and Marine Corps, and the rest to foreign customers.[6] The last U.S.-built F-4 went to South Korea, while the last F-4 built was an F-4EJ built by Mitsubishi Heavy Industries in Japan and delivered on 20 May 1981.[*citation needed*] As of 2008, 631 Phantoms were in service worldwide,[37] while the Phantoms were in use as a target drone (specifically QF-4Cs) operated by the U.S. military until 21 December 2016, when the Air Force officially ended use of the type.[38] As of June 2025, 98 Phantom IIs remained in service with Air Forces around the world. Greece has 17, Turkey 19 and Iran 62.[*citation needed*] The numbers in Iran have decreased after Israeli bombing during the 2026 war destroyed at least one F-4 at Iran's Tabriz Airbase, and many at Iran's 3rd tactical airbase in Hamadan Province.[39]

## World records

To show off their new fighter, the Navy led a series of record-breaking flights early in Phantom development:[6] All in all, the Phantom set 16 world records. Five of the speed records remained unbeaten until the F-15 Eagle appeared in 1975.[9]

Transcontinental "Operation LANA" in 1961

- **Operation Top Flight**: On 6 December 1959, the second XF4H-1 performed a zoom climb to a world record 98,557 ft (30,040 m).[10][40] Commander Lawrence E. Flint Jr., USN accelerated his aircraft to Mach 2.5 (2,660 km/h; 1,650 mph) at 47,000 ft (14,330 m) and climbed to 90,000 ft (27,430 m) at a 45° angle. He then shut down the engines and glided to the peak altitude. As the aircraft fell through 70,000 ft (21,300 m), Flint restarted the engines and resumed normal flight.[41]
- On 5 September 1960, an F4H-1 averaged 1,216.78 mph (1,958.16 km/h) over a 500 km (311 mi) closed-circuit course.[10]
- On 25 September 1960, an F4H-1F averaged 1,390.24 mph (2,237.37 km/h) over a 100 km (62.1 mi) closed-circuit course.[10] FAIRecord File Number 8898.
- **Operation LANA**: To celebrate the 50th anniversary of Naval aviation (L is the Roman numeral for 50 and ANA stood for Anniversary of Naval Aviation) on 24 May 1961, Phantoms flew across the continental United States in under three hours and included several tanker refuelings. The fastest of the aircraft averaged 869.74 mph (1,400.28 km/h) and completed the trip in 2 hours 47 minutes, earning the pilot (and future NASA Astronaut), Lieutenant[42] Richard Gordon, USN and RIO, Lieutenant Bobbie Young, USN, the 1961 Bendix trophy.[10][43][44][45]
- **Operation Sageburner**: On 28 August 1961, a F4H-1F Phantom II averaged 1,452.777 kilometers per hour (902.714 miles per hour) over a 3 mi (4.82 km) course flying below 125 feet (38.1 m) at all times.[10] Commander J.L. Felsman, USN was killed during the first attempt at this record on 18 May 1961 when his aircraft disintegrated in the air after pitch damper failure.[46]
- **Operation Skyburner**: On 22 November 1961, a modified Phantom with water injection, piloted by Lt. Col. Robert B. Robinson, set an absolute world record average speed over a 20-mile (32.2 km) long 2-way straight course of 1,606.342 mph (2,585.086 km/h).[10][47][48][49]
- On 5 December 1961, another Phantom set a sustained altitude record of 66,443.8 feet (20,252 m).[10]
- **Project High Jump**: A series of time-to-altitude records was set in early 1962: 34.523 seconds to 3,000 m (9,840 ft), 48.787 seconds to 6,000 m (19,700 ft), 61.629 seconds to 9,000 m (29,500 ft), 77.156 seconds to 12,000 m (39,400 ft), 114.548 seconds to 15,000 m (49,200 ft), 178.5 s to 20,000 m (65,600 ft), 230.44 s to 25,000 m (82,000 ft), and 371.43 s to 30,000 m (98,400 ft).[50] All High Jump records were set by F4H-1 production number 108 (Bureau Number 148423).[51][52][53][54] Two of the records were set by future distinguished NASA astronaut LCdr John Young.[55][56]

# Design

## Overview

The F-4 Phantom is a tandem-seat fighter-bomber designed as a carrier-based interceptor to fill the U.S. Navy's fleet defense fighter role. Innovations in the F-4 included use of pulse-doppler radar (only on late variants such as the F-4F) and extensive use of titanium in its airframe.[57]

Despite imposing dimensions and a maximum takeoff weight of over 60,000 lb (27,000 kg),[58] the F-4 has a top speed of Mach 2.23 and an initial climb rate of over 41,000 ft/min (210 m/s).[59] The F-4's nine external hardpoints have a capability of up to 18,650 pounds (8,480 kg) of weapons, including air-to-air and air-to-surface missiles, and unguided, guided, and thermonuclear weapons.[60] Like other interceptors of its day, the F-4 was designed without an internal cannon.[61]

The baseline performance of a Mach 2-class fighter with long range and a bomber-sized payload would be the template for the next generation of large and light/middle-weight fighters optimized for daylight air combat.[62]

Cockpit of F-4 Phantom II

### Flight characteristics [ edit ]

"Speed is life" was F-4 pilots' slogan, as the Phantom's greatest advantage in air combat was acceleration[24] and thrust, which permitted a skilled pilot to engage and disengage from the fight at will. MiGs usually could outturn the F-4 because of the high drag on the Phantom's airframe;[63] as a massive fighter aircraft designed to fire radar-guided missiles from beyond visual range, the F-4 lacked the agility of its Soviet opponents and was subject to adverse yaw during hard maneuvering. Although the F-4 was subject to irrecoverable spins during aileron rolls, pilots reported the aircraft to be very responsive and easy to fly on the edge of its performance envelope. In 1972, the F-4E model was upgraded with leading edge slats on the wing, greatly improving high angle of attack maneuverability at the expense of top speed.[64]

Cockpit of an F-4 Phantom II at the Estonian Aviation Museum

The J79 had a reduced time lag between the pilot advancing the throttle, from idle to maximum thrust, and the engine producing maximum thrust compared to earlier engines. While landing on USS *Midway* (CV-41) John Chesire's tailhook missed the arresting gear as he (mistakenly) reduced thrust to idle. He then slammed the throttle to full afterburner, the engine's response time being enough to return to full thrust quickly, and he was able get the Phantom airborne again successfully (bolter).[24] The J79 produced noticeable amounts of black smoke (at mid-throttle/cruise settings), a severe disadvantage in that it made it easier for the enemy to spot the aircraft.[65] Two decades after the aircraft entered service[24] this was solved on the F-4S, which was fitted with the −10A engine variant with a smokeless combustor.[66]

F-4 Phantom II flight demonstration video

The lack of an internal gun "was the biggest mistake on the F-4", Chesire said; "Bullets are cheap and tend to go where you aim them. I needed a gun, and I really wished I had one." Marine Corps General John R. Dailey recalled that "everyone in RF-4s wished they had a gun on the aircraft."[24] For a brief period, doctrine held that turning combat would be impossible at supersonic speeds and little effort was made to teach pilots air combat maneuvering. In reality, engagements quickly became subsonic, as pilots would slow down in an effort to get behind their adversaries. Furthermore, the relatively new heat-seeking and radar-guided missiles at the time were frequently reported as unreliable and pilots had to fire multiple missiles just to hit one enemy fighter. To compound the problem, rules of engagement in Vietnam precluded long-range missile attacks in most instances, as visual identification was normally required. Many pilots found themselves on the tail of an enemy aircraft, but too close to fire short-range Falcons or Sidewinders. Although by 1965 USAF F-4Cs began carrying SUU-16 external gunpods containing a 20 mm (.79 in) M61A1 Vulcan Gatling cannon, USAF cockpits were not equipped with lead-computing gunsights until the introduction of the SUU-23, virtually assuring a miss in a maneuvering fight. Some Marine Corps aircraft carried two pods for strafing. In addition to the loss of performance due to drag, combat showed the externally mounted cannon to be inaccurate unless frequently boresighted, yet far more cost-effective than missiles. The lack of a cannon was finally addressed by adding an internally mounted 20 mm (.79 in) M61A1 Vulcan on the F-4E.[64]

### Costs [ edit ]

| | F-4C | RF-4C | F-4D | F-4E |
|---|---|---|---|---|
| Unit R&D cost | – | 61,200 (1965) by 1973<br>625,248 (current) by 1973 | – | 22,700 (1965) by 1973<br>231,914 (current) by 1973 |
| Airframe | 1,388,725 (1965)<br>14,187,872 (current) | 1,679,000 (1965)<br>17,153,458 (current) | 1,018,682 (1965)<br>10,407,337 (current) | 1,662,000 (1965)<br>16,979,778 (current) |
| Engines | 317,647 (1965)<br>3,245,232 (current) | 276,000 (1965)<br>2,819,747 (current) | 260,563 (1965)<br>2,662,035 (current) | 393,000 (1965)<br>4,015,074 (current) |
| Electronics | 52,287 (1965)<br>534,189 (current) | 293,000 (1965)<br>2,993,427 (current) | 262,101 (1965)<br>2,677,748 (current) | 299,000 (1965)<br>3,054,725 (current) |
| Armament | 139,706 (1965)<br>1,427,303 (current) | 73,000 (1965)<br>745,803 (current) | 133,430 (1965)<br>1,363,184 (current) | 111,000 (1965)<br>1,134,029 (current) |
| Ordnance | – | – | 6,817 (1965)<br>69,646 (current) | 8,000 (1965)<br>81,732 (current) |
| Flyaway cost | 1.9 million (1965)<br>19.4 million (current) | 2.3 million (1965)<br>23.5 million (current) | 1.7 million (1965)<br>17.4 million (current) | 2.4 million (1965)<br>24.5 million (current) |
| Modification costs | 116,289 (1965) by 1973<br>1,188,063 (current) by 1973 | 55,217 (1965) by 1973<br>564,123 (2008) by 1973 | 233,458 (1965) by 1973<br>2,385,117 (current) by 1973 | 7,995 (1965) by 1973<br>81,681 (current) by 1973 |
| Cost per flying hour | 924 (1965)<br>9,440 (2008) | 867 (1965)<br>8,858 (current) | 896 (1965)<br>9,154 (current) | 867 (1965)<br>8,858 (current) |
| Maintenance cost per flying hour | 545 (1965)<br>5,568 (current) | | | |

Note: Original amounts were in 1965 U.S. dollars.[67] The figures in these tables have been adjusted for inflation to the current year.

## Operational history [ edit ]

*See also: McDonnell Douglas F-4 Phantom II U.S. operators*

### United States Navy [ edit ]

A U.S. Navy F-4B from VF-111 dropping bombs over Vietnam, 25 November 1971

On 30 December 1960, VF-121 *Pacemakers* at NAS Miramar became the first Phantom operator with its F4H-1Fs (F-4As). The VF-74 *Be-devilers* at NAS Oceana became the first deployable Phantom squadron when it received its F4H-1s (F-4Bs) on 8 July 1961.[68] The squadron completed carrier qualifications in October 1961 and Phantom's first full carrier deployment between August 1962 and March 1963 aboard *Forrestal*.[69] The second deployable U.S. Atlantic Fleet squadron to receive F-4Bs was the VF-102 *Diamondbacks*, who promptly took their new aircraft on the shakedown cruise of *Enterprise*.[70] The first deployable U.S. Pacific Fleet squadron to receive the F-4B was the VF-114 *Aardvarks*, which participated in the September 1962 cruise aboard USS *Kitty Hawk*.[68]

By the time of the Tonkin Gulf incident, 13 of 31 deployable navy squadrons were armed with the type. F-4Bs from *Constellation* made the first Phantom combat sortie of the Vietnam War on 5 August 1964, flying bomber escort in Operation Pierce Arrow.[71] Navy fighter pilots were unused to flying with a non-pilot RIO, but learned from air combat in Vietnam the benefits of the GiB "guy in back" or "voice in the luggage compartment" helping with the workload.[24] The first Phantom air-to-air victory of the war took place on 9 April 1965 when an F-4B from VF-96 *Fighting Falcons* piloted by Lieutenant (junior grade) Terence M. Murphy and his RIO, Ensign Ronald Fegan, shot down a Chinese MiG-17. The Phantom was then shot down, probably by an AIM-7 Sparrow from one of its wingmen.[26] There continues to be controversy over whether the Phantom was shot down by MiG guns or, as enemy reports later indicated, an AIM-7 Sparrow III from one of Murphy's and Fegan's wingmen.[72] On 17 June 1965, an F-4B from VF-21 *Freelancers* piloted by Commander Louis Page and Lieutenant John C. Smith shot down the first North Vietnamese MiG of the war.[73][74]

On 10 May 1972, Lieutenant Randy "Duke" Cunningham and Lieutenant (junior grade) William P. Driscoll flying an F-4J, call sign *Showtime 100*, shot down three MiG-17s to become the first American flying aces of the war. Their fifth victory was believed at the time to be over a mysterious North Vietnamese ace, Colonel Nguyen Toon, now considered mythical. On the return flight, the Phantom was damaged by an enemy surface-to-air missile. To avoid being captured, Cunningham and Driscoll flew their burning aircraft using only the rudder and afterburner (the damage to the aircraft rendered conventional control nearly impossible), until they could eject over water.[12]

The Blue Angels flew the F-4J from 1969 to 1974

During the war, U.S. Navy F-4 Phantom squadrons participated in 84 combat tours with F-4Bs, F-4Js, and F-4Ns. The Navy claimed 40 air-to-air victories at a cost of 73 Phantoms lost in combat (seven to enemy aircraft, 13 to SAMs and 53 to AAA). An additional 54 Phantoms were lost in mishaps.[75]

In 1984, all Navy F-4Ns were retired from Fleet service in deployable USN squadrons and by 1987 the last F-4Ss were retired from deployable USN squadrons. On 25 March 1986, an F-4S belonging to the VF-151 *Vigilantes*, became the last active duty U.S. Navy Phantom to launch from an aircraft carrier, in this case, *Midway*. On 18 October 1986, an F-4S from the VF-202 *Superheats*, a Naval Reserve fighter squadron, made the last-ever Phantom carrier landing while operating aboard *America*. In 1987, the last of the Naval Reserve-operated F-4S aircraft were replaced by F-14As. The last Phantoms in service with the Navy were QF-4N and QF-4S target drones operated by the Naval Air Warfare Center at NAS Point Mugu, California.[26] These were subsequently retired in 2004.[76]

### United States Marine Corps [ edit ]

A U.S. Marine F-4B with VMFA-314, flies over South Vietnam in September 1968

The Marine Corps received its first F-4Bs in June 1962, with the *Black Knights* of VMFA-314 at Marine Corps Air Station El Toro, California becoming the first operational squadron. Marine Phantoms of VMFA-323 *Death Rattlers*, flying from Puerto Rico, provided air cover during Operation Power Pack for the evacuation of US citizens from the Dominican Republic[77] and assisted the 508th Infantry Regiment in taking and securing a position east of the Duarte bridge. Marine Phantoms from VMFA-531 *Grey Ghosts* were assigned to Da Nang Air Base on South Vietnam's northeast coast on 10 May 1965 and were initially assigned to provide air defense for the USMC. They soon began close air support missions (CAS) and VMFA-314, VMFA-232 *Red Devils*, VMFA-323 *Death Rattlers* and VMFA-542 *Tigers* joined them.[78]

Marine Phantoms from VMFA-323 and VMFA-531 operating from the USS *Coral Sea* participated in Operation Eagle Claw, the attempted rescue of American hostages from Iran, with orders to shoot down any Iranian aircraft. The Phantoms were painted with an orange stripe enclosed by two black stripes in order to distinguish the American F-4s from the Iranian F-4s.[79][80] The operation was called off in the early stages of execution.

The VMCJ-1 *Golden Hawks* (later VMAQ-1 and VMAQ-4 which had the old RM tailcode) flew the first photo recon mission with an RF-4B variant on 3 November 1966 from Da Nang and remained there until 1970 with no RF-4B losses and only one aircraft damaged by anti-aircraft artillery (AAA) fire.[81] VMCJ-2 and VMCJ-3 (now VMAQ-3) provided aircraft for VMCJ-1 in Da Nang and VMFP-3 was formed in 1975 at MCAS El Toro, CA consolidating all USMC RF-4Bs in one unit that became known as "The Eyes of the Corps." VMFP-3 disestablished in August 1990 after the Advanced Tactical Airborne Reconnaissance System was introduced for the F/A-18D Hornet.[29]

The F-4 continued to equip fighter-attack squadrons in both active and reserve Marine Corps units throughout the 1960s, 1970s and 1980s and into the early 1990s. In the early 1980s, these squadrons began to transition to the F/A-18 Hornet, starting with the same squadron that introduced the F-4 to the Marine Corps, VMFA-314 at MCAS El Toro, California. On 18 January 1992, the last Marine Corps Phantom, an F-4S in the Marine Corps Reserve, was retired by the *Cowboys* of VMFA-112 at NAS Dallas, Texas, after which the squadron was re-equipped with F/A-18 Hornets.[82]

### United States Air Force [ edit ]

**USAF F-4 Summary for Vietnam War action**

| Aircraft | Weapons/Tactics | MiG-17 | MiG-19 | MiG-21 | Total |
|---|---|---|---|---|---|
| F-4C | AIM-7 Sparrow | 4 | 0 | 10 | 14 |
| | AIM-9 Sidewinder | 12 | 0 | 10 | 22 |
| | 20 mm gunpod | 3 | 0 | 1 | 4 |
| | Maneuvering tactics | 2 | 0 | 0 | 2 |
| F-4D | AIM-4 Falcon | 4 | 0 | 1 | 5 |
| | AIM-7 Sparrow | 4 | 2 | 20 | 26 |
| | AIM-9 Sidewinder | 0 | 2 | 3 | 5 |
| | 20 mm gunpod | 4.5 | 0 | 2 | 6.5 |
| | Maneuvering tactics | 0 | 0 | 2 | 2 |
| F-4E | AIM-7 Sparrow | 0 | 2 | 8 | 10 |
| | AIM-9 Sidewinder | 0 | 0 | 4 | 4 |
| | AIM-9 and 20 mm gunpod | 0 | 0 | 1 | 1 |
| | 20 mm gunpod | 0 | 1 | 4 | 5 |
| | Maneuvering tactics | 0 | 1 | 0 | 1 |
| Total | | 33.5 | 8 | 66 | 107.5 |

In USAF service, the F-4 was initially designated the F-110A[83] prior to the introduction of the 1962 United States Tri-Service aircraft designation system. The USAF quickly embraced the design and became the largest Phantom user. The first Phantoms that the USAF operated were F-4Bs loaned from the Navy, with 27 jets delivered to the 4453rd Combat Crew Training Wing at MacDill Air Force Base, Florida, in November 1963. The first operational unit was the 12th Tactical Fighter Wing, who received the USAF's first F-4Cs in January 1964, achieving initial operational capability (IOC) in October 1964.[84] The first USAF Phantoms to participate in the Vietnam War were F-4Cs from the 45th Tactical Fighter Squadron, who deployed to Ubon Royal Thai Air Force Base, Thailand, in April 1965.[84][85][86]

Unlike the U.S. Navy and U.S. Marine Corps, which flew the Phantom with a Naval Aviator (pilot) in the front seat and a naval flight officer as a radar intercept officer (RIO) in the back seat, the USAF initially flew its Phantoms with a rated Air Force Pilot in front and back seats. Pilots usually did not like flying in the back seat;[24] while the GIB, or "guy in back", could fly and ostensibly land the aircraft, they had fewer flight instruments and a very restricted forward view. The Air Force later assigned a rated Air Force Navigator qualified as a weapon/targeting systems officer (later designated as weapon systems officer or WSO) in the rear seat instead of another pilot.[87][24]

On 10 July 1965, F-4Cs of the 45th TFS, 15th TFW,[88] scored the USAF's first victories against North Vietnamese MiG-17s using AIM-9 Sidewinder air-to-air missiles.[89] On 26 April 1966, an F-4C from the 480th Tactical Fighter Squadron scored the first aerial victory by a U.S. aircrew over a North Vietnamese MiG-21 "Fishbed".[90] On 24 July 1965, another Phantom from the 45th Tactical Fighter Squadron became the first American aircraft to be downed by an enemy SAM, and on 5 October 1966 an 8th Tactical Fighter Wing F-4C became the first U.S. jet lost to an air-to-air missile, fired by a MiG-21.

On 2 January 1967, F-4Cs of the 8th Tactical Fighter Wing, under the command of Robin Olds, executed Operation Bolo, a response to heavy losses sustained during Operation Rolling Thunder. Olds' and his flight flew out of Ubon in Thailand and simulated an F-105 strike force. In response, the VPAF sent up MiG-21s to shoot down the Phantoms. The ensuing battle resulted in the VPAF losing half of their MiG-21 fleet with no losses from the American side.

Early aircraft suffered from leaks in wing fuel tanks that required re-sealing after each flight and 85 aircraft were found to have cracks in outer wing ribs and stringers.[67] There were also problems with aileron control cylinders, electrical connectors, and engine compartment fires. Reconnaissance RF-4Cs made their debut in Vietnam on 30 October 1965, flying the hazardous post-strike reconnaissance missions. The USAF Thunderbirds used the F-4E from the 1969 season until 1974.[15]

435th TFS F-4Ds over Vietnam

Although the F-4C was essentially identical to the Navy/Marine Corps F-4B in-flight performance and carried the AIM-9 Sidewinder missiles, USAF-tailored F-4Ds initially arrived in June 1967 equipped with AIM-4 Falcons. However, the Falcon, like its predecessors, was designed to shoot down heavy bombers flying straight and level. Its reliability proved no better than others and its complex firing sequence and limited seeker-head cooling time made it virtually useless in combat against agile fighters. The F-4Ds reverted to using Sidewinders under the "Rivet Haste" program in early 1968, and by 1972 the AIM-7E-2 "Dogfight Sparrow" had become the preferred missile for USAF pilots. Like other Vietnam War Phantoms, the F-4Ds were urgently fitted with radar warning receivers to detect the Soviet-built S-75 Dvina SAMs.[91]

From the initial deployment of the F-4C to Southeast Asia, USAF Phantoms performed both air superiority and ground attack roles, supporting not only ground troops in South Vietnam, but also conducting bombing sorties in Laos and North Vietnam. As the F-105 force underwent severe attrition between 1965 and 1968, the bombing role of the F-4 proportionately increased until after November 1970 (when the last F-105D was withdrawn from combat) it became the primary USAF tactical ordnance delivery system. In October 1972 the first squadron of EF-4C Wild Weasel aircraft deployed to Thailand on temporary duty.[92] The "E" prefix was later dropped and the aircraft was simply known as the F-4C Wild Weasel.

Sixteen squadrons of Phantoms were permanently deployed to Indochina between 1965 and 1973, and 17 others deployed on temporary combat assignments.[93] Peak numbers of combat F-4s occurred in 1972, when 353 were based in Thailand.[94] A total of 445 Air Force Phantom fighter-bombers were lost, 370 in combat and 193 of those over North Vietnam (33 to MiGs, 30 to SAMs and 307 to AAA).[94]

The RF-4C was operated by four squadrons,[95] and of the 83 losses, 72 were in combat including 38 over North Vietnam (seven to SAMs and 65 to AAA).[94] By war's end, the U.S. Air Force had lost a total of 528 F-4 and RF-4C Phantoms. When combined with U.S. Navy and Marine Corps losses of 233 Phantoms, 761 F-4/RF-4 Phantoms were lost in the Vietnam War.[96]

McDonnell Douglas F-4E "Phantom II"
USAF Museum Photo Archives
321
Battle-damaged F-4E Phantom II (S/N 67-321) of the 366th TFW, recovered by 1st Lts. Zimmerman and Craighead after MiG, SAM, and AAA damage. Despite losing hydraulics, rudder, and slats, Zimmerman landed successfully. (U.S. Air Force photo)

On 28 August 1972, Captain Steve Ritchie became the first USAF ace of the war.[11] On 9 September 1972, WSO Capt Charles B. DeBellevue became the highest-scoring American ace of the war with six victories.[11] and WSO Capt Jeffrey Feinstein became the last USAF ace of the war on 13 October 1972.[97] Upon return to the United States, DeBellevue and Feinstein were assigned to undergraduate pilot training (Feinstein was given a vision waiver) and requalified as USAF pilots in the F-4. USAF F-4C/D/E crews claimed 107.5 MiG kills in Southeast Asia (50 by Sparrow, 31 by Sidewinder, five by Falcon, 15.5 by gun, and six by other means).[94]

Gun-camera recorded kill from F-4 Phantom II piloted by Captains Steve Ritchie & Jeff Feinstein.

On 31 January 1972, the 170th Tactical Fighter Squadron, 183d Tactical Fighter Group of the Illinois Air National Guard became the first Air National Guard (ANG) unit to transition to Phantoms from Republic F-84F Thunderstreaks.[98] Phantoms would eventually equip numerous tactical fighter and tactical reconnaissance units in the USAF active, ANG, and Air Force Reserve (AFRES).[99]

On 2 June 1972, a Phantom flying at supersonic speed shot down a MiG-19 over Thud Ridge in Vietnam with its cannon. At a recorded speed of Mach 1.2, Major Phil Handley's shoot down was the first and only recorded gun kill while flying at supersonic speeds.[100][101]

In early December 1989, USAF F-4s, from Clark Air Base, participated in Operation Classic Resolve,[102] President George H. W. Bush's response to the 1989 Philippine coup attempt. The F-4s were ordered to buzz the rebel planes at their base, fire at them if any tried to take off, and shoot them down if they did. The buzzing by the US F-4s soon caused the coup to collapse. On 2 December, President Bush reported that on 1 December, US fighter aircraft from Clark Air Base assisted Philippine President Corazon Aquino's government repel a coup attempt.[103]

On 15 August 1990, 24 F-4G Wild Weasel Vs and six RF-4Cs were deployed to Isa Air Base, Bahrain, for Operation Desert Storm. The F-4G was the only aircraft in the USAF inventory equipped for the Suppression of Enemy Air Defenses role, and was needed to protect coalition aircraft from Iraq's extensive air defense system. The RF-4C was the only aircraft equipped with the ultra-long-range KS-127 LOROP (long-range oblique photography) camera and was used for a variety of reconnaissance missions. In spite of flying almost daily missions, only one RF-4C was lost in a fatal accident before the start of hostilities. One F-4G was lost when enemy fire damaged the fuel tanks and the aircraft ran out of fuel near a friendly airbase. The last USAF Phantoms, F-4G Wild Weasel Vs from 561st Fighter Squadron, were retired on 26 March 1996. The last operational flight of the F-4G Wild Weasel was from the 190th Fighter Squadron, Idaho Air National Guard, in April 1996.[104]

F-4Gs over Bahrain during Operation Desert Shield

### Target drone [ edit ]

Like the Navy, the Air Force also operated QF-4 target drones, serving with the 82d Aerial Targets Squadron at Tyndall Air Force Base, Florida, and Holloman Air Force Base, New Mexico.[105] Replacing the QF-106, the QF-4 program achieved IOC in 1997,[106] with the last QF-106 being shot down on 20 February 1997.[107] It was expected that the F-4 would remain in the target role with the 82d ATRS until at least 2015, when they would be replaced by early versions of the F-16 Fighting Falcon converted to a QF-16 configuration.[108]

On 19 November 2013, BAE Systems delivered its 314th, and last, QF-4 aerial target to the Air Force.[109] The example, RF-4C *68-0599*, had been in storage for over 20 years before being converted.[110] Over 16 years, BAE had converted 314 F-4 and RF-4 Phantom IIs into QF-4s and QRF-4s, with each aircraft taking six months to adapt.[109] By December 2013, QF-4 and QRF-4 aircraft had flown over 16,000 manned and 600 unmanned training sorties, with 250 unmanned aircraft being shot down in firing exercises.[109] The remaining QF-4s and QRF-4s held their training role until the first of 126 QF-16s were delivered by Boeing.[109] The first QF-16 was delivered to Tyndall AFB in September 2014, achieving IOC on 23 September 2015.[111] The final flight of an Air Force QF-4 from Tyndall AFB took place on 27 May 2015 to Holloman AFB.[112]

QF-4E flying over the Gulf of Mexico, 1998

After Tyndall AFB ceased operations, the 53d Weapons Evaluation Group at Holloman became the fleet of 22 QF-4s' last remaining operator. The base continued using them to fly manned test and unmanned live fire test support and Foreign Military Sales testing.[113] The final unmanned flight took place on 17 August 2016, with QF-4E *72-0166* being fired upon by a Lockheed Martin F-35 Lightning II before returning safely back to Holloman AFB.[114] The type was officially retired from US military service with a four–ship flight at Holloman during an event on 21 December 2016.[115] The last 13 QF-4s were stripped of their systems after 1 January 2017 and transferred to the White Sands Missile Range to be used as static targets.[111][116] During its career as a target drone, several QF-4s retained the capability to be flown as a manned aircraft and were maintained in historical color schemes, being displayed as part of Air Combat Command's Heritage Flight at air shows, base open houses, and other events while serving as non-expendable target aircraft during the week.[117]

## Aerial combat in the Vietnam War [ edit ]

The USAF and the US Navy had high expectations of the F-4 Phantom, assuming that the massive firepower, the best available on-board radar, the highest speed and acceleration properties, coupled with new tactics, would provide Phantoms with an advantage over the MiGs. However, in confrontations with the lighter MiG-21, F-4s did not always succeed and began to suffer losses.[118] Over the course of the air war in Vietnam, between 3 April 1965 and 8 January 1973, each side would ultimately claim favorable kill ratios.[119]

During the war, U.S. Navy F-4 Phantoms claimed 40 air-to-air victories at a loss of seven Phantoms to enemy aircraft.[75] USMC F-4 pilots claimed three enemy MiGs at the cost of one aircraft in air-combat. USAF F-4 Phantom crews scored $107\frac{1}{2}$ MiG kills (including $33\frac{1}{2}$ MiG-17s, eight MiG-19s and 66 MiG-21s) at a cost of 33 Phantoms in air-combat.[94] F-4 pilots were credited with a total of $150\frac{1}{2}$ MiG kills at a cost of 42 Phantoms in air-combat.

According to the Vietnam People's Air Force (VPAF), 103 F-4 Phantoms were shot down by MiG-21s at a cost of 54 MiG-21s downed by F-4s.[120] During the war, the VPAF lost 131 MiGs in air combat (63 MiG-17s, eight MiG-19s and 60 MiG-21s) of which one half were by F-4s.[121]

From 1966 to November 1968, in 46 air battles conducted over North Vietnam between F-4s and MiG-21s, VPAF claimed 27 F-4s were shot down by MiG-21s at a cost of 20 MiG-21s[122] In 1970, one F-4 Phantom was shot down by a MiG-21.[123] The struggle culminated on 10 May 1972, with VPAF aircraft completing 64 sorties, resulting in 15 air battles. The VPAF claimed seven F-4s were shot down, while U.S. confirmed five F-4s were lost.[123] The Phantoms, in turn, managed to destroy two MiG-21s, three MiG-17s, and one MiG-19.[122] On 11 May, two MiG-21s, which played the role of "bait", brought the four F-4s to two MiG-21s circling at low altitude. The MiGs quickly engaged and shot down two F-4s. On 18 May, Vietnamese aircraft made 26 sorties in eight air engagements, which cost 4 F-4 Phantoms; Vietnamese fighters on that day did not suffer losses.[122]

On 5 August 1967, the USS *Forrestal* was stationed off the Indochina coast to carry out strikes against North Vietnam. An electrical fault caused a Zuni rocket to be fired from an F-4. The rocket struck the fuel tank of an A-4 Skyhawk attack aircraft, starting a fire which quickly spread to other airplanes, setting off several bombs. The fire and explosions killed 134 men and seriously wounded 161 more in what became known as the 1967 USS Forrestal fire.[124][125]

## Non-U.S. users [ edit ]

*Main article: McDonnell Douglas F-4 Phantom II non-U.S. operators*

The Phantom has served with the air forces of many countries, including Australia, Egypt, Germany, United Kingdom, Greece, Iran, Israel, Japan, Spain, South Korea and Turkey.

### Australia [ edit ]

*Main article: McDonnell Douglas F-4 Phantom II in Australian service*

The Royal Australian Air Force (RAAF) leased 24 USAF F-4Es from 1970 to 1973 while waiting for their order for the General Dynamics F-111C to be delivered. They were so well-liked that the RAAF considered retaining the aircraft after the F-111Cs were delivered.[126] They were operated from RAAF Amberley by No. 1 Squadron and No. 6 Squadron.[127]

### Egypt [ edit ]

In 1979, the Egyptian Air Force purchased 35 former USAF F-4Es along with a number of Sparrow, Sidewinder and Maverick missiles from the U.S. for $594 million as part of the "Peace Pharaoh" program.[128] An additional seven surplus USAF aircraft were purchased in 1988.[129] Three attrition replacements had been received by the end of the 1990s.[126]

Egyptian F-4Es were retired in 2020, with their former base at Cairo West Air Base being reconfigured for the operation of F-16C/D Fighting Falcons.[130]

### Germany [ edit ]

F-4Fs of the German Air Force, 21 January 1998

The West German Air Force (*Luftwaffe*) found itself in need of numerous new aircraft following the December 1967 publication of NATO's new flexible response doctrine. With the move back towards fighting a conventional war in Europe, there was a need for both far more capable photoreconnaissance and fighter aircraft, with the RF-104G and F-104G fleets respectively being considered obsolescent due to their inability to operate in bad weather, at night, and in the latter's case, its performance being deemed insufficient for the air defense and air superiority missions against the MiG-21 that was fielded en masse by Warsaw Pact air forces. These new aircraft would need to enter service relatively quickly as well, with age and fatigue rates putting Luftwaffe strength below the required numbers by 1976. However, with the ongoing development of the Panavia Tornado there was insufficient capital to design new aircraft for these roles, and thus foreign models would have to be procured.[131]

#### Reconnaissance role [ edit ]

The photoreconnaissance aircraft selection pool was relatively small to start with, consisting of the Canadair CF-5A(R), Mirage IIIR, Saab S 35E, Lockheed RTF-104G, and McDonnell Douglas RF-4E. The competition came down to the RTF-104G and RF-4E, the former being a heavily modified photoreconnaissance variant of the TF-104G and the latter an F-4E based equivalent to the USAF's own RF-4C intended for export customers. As the RTF-104G could still reuse most existing infrastructure at bases operating RF-104Gs while RF-4Es could not, the latter had a substantially higher unit price - 23 million Deutsche Marks for an RF-4E compared to 8 million DM for an RTF-104G. The latter was also considered slightly more modern, and could be built under license on the existing F-104G production line at Messerschmitt-Bölkow-Blohm (MBB), meaning most of the money would be kept in West Germany. However, the safety and power advantages of the Phantom were viewed very favorably - the USAF had experienced loss rates of 9 F-4 versus 25 F-104 for every 100,000 flight hours, and West German experience with the F-104 platform was marred with crashes. Another factor in the Phantom's favor was industrial offsets, as West Germany had set aside 3 billion DM to purchase defense equipment from the U.S.[132] which could be used outside of the normal defense budget to fund this acquisition. In 1969, the German Defense Ministry decided to procure 88 RF-4Es, with the contract costing 2.052 billion DM. While not manufactured in West Germany, their construction was not wholly American. Companies like MBB produced stabilizers, landing gear doors, outer wings, and ailerons which were then shipped to McDonnell Douglas's St. Louis factory for final assembly.[133]

The RF-4E first flew on 15 September 1970, and the West Germans took delivery of their first four aircraft at Bremgarten AB on 20 January 1971.[134] To act as flight instructors, nine pilots and nine navigators were sent to Shaw AFB between 22 January 1970 and 5 May 1971 to attend eight to nine month long training courses on the RF-4C. 108 pilots were trained at George AFB between 10 October 1970 and 4 February 1972 for five week flight training courses. These pilots had to receive sensor training from the aforementioned flight instructors after returning to their units in West Germany.[135]

With the lethality of Warsaw Pact air defenses increasing along with the strength of their conventional ground forces over the course of the 1970s, in 1978 it was decided to simultaneously retrofit the aircraft for a secondary ground attack role using the same equipment as the F-4F, and to improve their self-defense capabilities by fitting improved flare and chaff systems and a more capable radar warning receiver. New infrared cameras were also fitted for improved reconnaissance capabilities.[136] 71 aircraft were again upgraded in the mid 1980's, as planners saw no replacement for the type available before 2005. These included an increase in maximum flight hours for 4,500 to 8,000 per airframe, the installation of a laser and GPS navigation system, a new AN/APQ-172 terrain following radar, and integration for the new AIM-9L Sidewinder.[137] However, after German reunification and the end of the Cold War, there was a sharp drawdown in military spending.[138] 27 RF-4Es were sold to the Greek Air Force, of which seven were spares. Another 46 were sold to the Turkish Air Force - 33 operational aircraft and 13 spares.[139] The Luftwaffe retired the type in 1994,[140] and the last example was handed over to the Museum of the Luftwaffe on 23 November 2003.[141]

#### Fighter role [ edit ]

50 MILES
40
30
20
10
60° 30° 0° 30° 60°
25:42

*F-4 Air Combat Tactics* (1968) De-classified official USAF aerial combat engagement pilot training film reel.

The fighter contract was designed to prevent the loss rates that had plagued F-104 in Luftwaffe service from repeating. Two of the key requirements in the new fighter program were an all-weather navigation system and two engines. The competition also featured a greater variety of contestants, with the SEPECAT Jaguar, Saab Viggen, Dassault Mirage F1, Northrop F-5, Northrop P-530, and McDonnell Douglas F-4F all being offered. With RF-4E already being adopted, combined with F-4F's advantages in range and weapons load, it was the declared the winner and an order for 175 aircraft placed on 24 June 1971 under the "Peace Rhine" program, with a unit price of approximately 12 million DM. The "F" variant was a simplified version of the "E", designed for maximum compatibility with RF-4E. The aircraft was roughly 1,500 kilograms (3,300 lb) lighter, lacking the ram air turbine, rear fuselage fuel tank, slatted stabilizers, and ability to use AIM-7 Sparrow.[142] However, these weight reductions combined with the leading edge slats that were not present on RF-4E made F-4F markedly more maneuverable, especially at low speeds. All but the first F-4F incorporated West German built subcomponents in the same manner as their RF-4Es.[143] All F-4Fs were delivered between 1973 and 1976.[144] These purchases made Germany the largest export customer for the Phantom.[145]

The F-4F first flew on 18 March 1973, and was publicly unveiled on 24 May of the same year. Shortly thereafter, the first 12 aircraft were delivered to George AFB in cooperation with the 35th Tactical Fighter Wing to stand up an operational conversion unit, with pilots of *Jagdgeschwader 71* (Fighter Wing 71) arriving on 1 January 1974 to begin training. Due to the urgent need for the aircraft in frontline service, West Germany replaced these F-4Fs with 10 new F-4Es in 1975, which were permanently stationed in the U.S. for training until their retirement in 1997.[146] A single F-4F was kept in the U.S. on loan to U.S. Air Force Systems Command until 1982 under the designation NTF-4F, which was used as a testbed for new technologies in the aircraft.[147] The F-4F was upgraded in the mid-1980s[148] to use more capable ordnance that was being introduced by the U.S. in light of new capabilities introduced by the Warsaw Pact militaries. These alterations included a new digital weapons computer, altering the radar to allow for bombing through cloud cover, a HUD, and integration of the AIM-9L Sidewinder and AGM-65B Maverick.[149] The droptank was replaced with a lower drag model used by F-15C, and both the radar warning receiver and jamming pods were upgraded. The chaff/flare dispensers were also replaced with more capable models.[150]

Much as the Warsaw Pact's improving conventional strengths had seen a need for a more capable reconnaissance aircraft, the increasing lethality of air defenses and aircraft saw a similar need for more capable fighters. While procurement of the Tornado ADV was considered, it was decided against in favor of upgrading the F-4F fleet. The program, known as Improved Combat Efficiency (ICE) or Kampfwertsteigerung (KWS) began in 1983 and sought to fit the aircraft with substantially improved air to air and air to ground weapons, including beyond-visual-range missiles and standoff weapons[151] To start, the 153 F-4Fs in frontline service all received modest upgrades known as KWS-LA (ground attack, luftangriff), including a laser navigation system and structural upgrades to extend maximum flight hours from 4,000 to 6,500 per airframe. 110 of these were further upgraded under KWS-LV (air defense, luftverteidigung) with the AN/APG-65GY radar, a new mission computer, and compatibility with AIM-120 AMRAAM.[152] The KWS-LV F-4Fs entered service in 1992,[148] and were expected to remain in service until 2012.[153] All the remaining Luftwaffe Phantoms were based at Wittmund with *Jagdgeschwader* 71 in Northern Germany[154] and WTD61 at Manching. A total of 24 F-4Fs were operated by the 49th Tactical Fighter Wing of the USAF at Holloman AFB to train Luftwaffe crews until December 2004. Phantoms were deployed to NATO states under the Baltic Air Policing starting in 2005, 2008, 2009, 2011 and 2012. The German Air Force retired its last F-4Fs on 29 June 2013. German F-4Fs flew 279,000 hours from entering service on 31 August 1973 until retirement.[155][156]

### Greece [ edit ]

In 1972, the Hellenic Air Force signed a contract for 36 brand new F-4E Phantoms, with deliveries starting in 1974.[157][158][159] In the early 1990s, the Hellenic AF acquired surplus RF-4Es and F-4Es from the *Luftwaffe* and U.S. ANG.[160][161] Following the success of the German ICE program, on 11 August 1997, a contract was signed between DASA of Germany and Hellenic Aerospace Industry for the upgrade of 39 aircraft to the very similar "Peace Icarus 2000" standard.[26] On 5 May 2017, the Hellenic Air Force officially retired the RF-4E Phantom II during a public ceremony.[162]

### Iran [ edit ]

Iranian F-4E Phantom refueling through a boom during Iran-Iraq war, 1982

In the 1960s and 1970s when the U.S. and Iran were on friendly terms, the U.S. delivered 225 F-4D, F-4E, and RF-4E Phantoms to Iran, making it the second largest export customer.[163] The Imperial Iranian Air Force saw at least one engagement, resulting in a loss, after an RF-4C was rammed[164] by a Soviet MiG-21 during Project Dark Gene, an ELINT operation during the Cold War.

The Islamic Republic of Iran Air Force Phantoms saw heavy action in the Iran–Iraq War in the 1980s and were kept operational by overhaul and servicing from Iran's aerospace industry.[165] Notable operations of Iranian F-4s during the war included Operation Scorch Sword, an attack by two F-4s against the Iraqi Osirak nuclear reactor site near Baghdad on 30 September 1980,[166] and the attack on H3, a 4 April 1981 strike by eight Iranian F-4s against the H-3 complex of air bases in the far west of Iraq, which resulted in many Iraqi aircraft being destroyed or damaged for no Iranian losses.[167]

On 5 June 1984, two Saudi Arabian fighter pilots shot down two Iranian F-4 fighters. The Royal Saudi Air Force pilots were flying American-built F-15s and fired air-to-air missiles to bring down the Iranian planes. The Saudi fighter pilots had Boeing KC-135 Stratotanker planes and Boeing E-3 Sentry AWACS surveillance planes assist in the encounter. The aerial fight occurred in Saudi airspace over the Persian Gulf near the Saudi island Al Arabiyah, about 60 miles northeast of Jubail.[168]

Iranian F-4s were in use as of late 2014;[169] the aircraft reportedly conducted air strikes on ISIS targets in the eastern Iraqi province of Diyala.[170] At the 2024 Kish Air Show, three F-4Es from the 91st Tactical Combat Squadron, operating out of their home base in Bandar Abbas, performed formation and solo flyovers.[171]

On 1 March 2026, F-4s were targeted by Israeli and American forces during the 2026 Iran war, with at least one F-4 claimed destroyed in an air strike in Tabriz.[172]

### Israel [ edit ]

Israeli Air Force F-4Es taking off during the Yom Kippur War in October 1973

The Israeli Air Force acquired between 212 and 222[173] newly built and ex-USAF aircraft, and modified several as one-off special reconnaissance variants. The first F-4Es, nicknamed "*Kurnass*" (Sledgehammer), and RF-4Es, nicknamed "*Orev*" (Raven), were delivered in 1969 under the "Peace Echo I" program. Additional Phantoms arrived during the 1970s under "Peace Echo II" through "Peace Echo V" and "Nickel Grass" programs. Israeli Phantoms saw extensive combat during Arab–Israeli conflicts, first seeing action during the War of Attrition.[174] In the 1980s, Israel began the "Kurnass 2000" modernization program which significantly updated avionics.[26] The last Israeli F-4s were retired in 2004.[175]

### Japan [ edit ]

From 1968, the Japan Air Self-Defense Force (JASDF) purchased a total of 140 F-4EJ Phantoms without aerial refueling, AGM-12 Bullpup missile system, nuclear control system or ground attack capabilities.[176][177] Mitsubishi built 138 under license in Japan and 14 unarmed reconnaissance RF-4Es were imported. One of the aircraft (*17-8440*) was the last of the 5,195 F-4 Phantoms to be produced. It was manufactured by Mitsubishi Heavy Industries on 21 May 1981. "The Final Phantom" served with 306th Tactical Fighter Squadron and later transferred to the 301st Tactical Fighter Squadron.[178]

JASDF F-4EJ *Kais* (*57-8354* and *87-8407*) of 8 Hikōtai in gray air superiority paint scheme, 2002

Of these, 96 F-4EJs were modified to the F-4EJ Kai (改; modified) standard.[179] 15 F-4EJ and F-4EJ *Kai* were converted to reconnaissance aircraft designated RF-4EJ. Japan had a fleet of 90 F-4s in service in 2007. After studying several replacement fighters[180][181] the F-35A Lightning II was chosen in 2011.[182] The 302nd Tactical Fighter Squadron became the first JASDF F-35 Squadron at Misawa Air Base when it converted from the F-4EJ *Kai* on 29 March 2019.[183] The JASDF's sole aerial reconnaissance unit, the 501st Tactical Reconnaissance Squadron, retired their RF-4Es and RF-4EJs on 9 March 2020, and the unit itself dissolved on 26 March.[184]

The 301st Tactical Fighter Squadron then became the sole user of the F-4EJ in the Air Defense Command, with their retirement originally scheduled in 2021 along with the unit's transition to the F-35A.[185] However, on 20 November 2020, the 301st Tactical Fighter Squadron announced the earlier retirement of their remaining F-4EJs, concluding the Phantom's long-running career in the JASDF Air Defense Command.[186] Although retirement was announced, the 301st TFS continued operations up until 10 December 2020, with the squadron's Phantoms being decommissioned on 14 December.[187] Two F-4EJs and a F-4EJ *Kai* continued to be operated by the Air Development and Test Wing in Gifu Prefecture until their retirement on 17 March 2021, marking an end of Phantom operations in Japan.[2]

### South Korea [ edit ]

The Republic of Korea Air Force received its first batch of used USAF F-4D Phantoms in 1969 under the "Peace Spectator" program. The F-4Ds continued to be delivered until 1988. The "Peace Pheasant II" program also provided new-built and former USAF F-4Es.[188] In 1975, South Korea fundraised to buy five new F-4 Phantoms with the money donated from South Korean citizens in a national fundraising drive set up to buy the jets (a necessity at the time as South Korea was a poor country and in the aftermath of the Korean War, needed the extra jets to defend against North Korea which had a larger air force back then).[189] In total, the ROKAF operated 92 F-4Ds, 27 RF-4Cs, and 103 F-4Es.[190] The last ROKAF F-4Es were retired on 7 June 2024.[3]

### Spain [ edit ]

The Spanish Air Force acquired its first batch of ex-USAF F-4C Phantoms in 1971 under the "Peace Alfa" program. Designated C.12, the aircraft were retired in 1989. At the same time, the air arm received a number of ex-USAF RF-4Cs, designated CR.12. In 1995–1996, these aircraft received extensive avionics upgrades. Spain retired its RF-4s in 2002.[191][192]

#### Turkey [ edit ]

A Turkish Air Force F-4E Phantom II aircraft armed with Popeye missiles takes off from Third Air Force Base Konya, Turkey, during Exercise Anatolian Eagle.

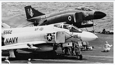
The Turkish Air Force (TAF) received 40 F-4Es in 1974, with a further 32 F-4Es and 8 RF-4Es in 1977–78 under the "Peace Diamond III" program, followed by 40 ex-USAF aircraft in "Peace Diamond IV" in 1987, and a further 40 ex-U.S. Air National Guard Aircraft in 1991.[193] A further 32 RF-4Es were transferred to Turkey after being retired by the Luftwaffe between 1992 and 1994.[193] In 1995, Israel Aerospace Industries (IAI) implemented an upgrade similar to Kurnass 2000 on 54 Turkish F-4Es which were dubbed the F-4E 2020 Terminator.[26] Turkish F-4s, and more modern F-16s have been used to strike Kurdish PKK bases in ongoing military operations in Northern Iraq.[194] On 22 June 2012, a Turkish RF-4E was shot down by Syrian air defenses while flying a reconnaissance flight near the Turkish-Syrian border.[195][196] Turkey has stated the reconnaissance aircraft was in international airspace when it was shot down, while Syrian authorities stated it was inside Syrian airspace.[197] Turkish F-4s remained in use as of 2020,[169][198] and it plans to fly them at least until 2030.[199]

F-4E Phantom special livery for the 50th flight year of its service in the Turkish Air Force

On 24 February 2015, two RF-4Es crashed in the Malatya region in southeastern Turkey under unknown circumstances, killing both crewmembers of each aircraft.[200][201][202] On 5 March 2015, an F-4E-2020 crashed in central Anatolia killing both crew.[203][204] After the recent accidents, the TAF withdrew RF-4Es from active service. Turkey was reported to have used F-4 jets to attack PKK separatists and the ISIS capital on 19 September 2015.[205] The Turkish Air Force has reportedly used the F-4E 2020s against the more recent Third Phase of the PKK conflict on heavy bombardment missions into Iraq on 15 November 2015, 12 January 2016, and 12 March 2016.[206][207]

#### United Kingdom [ edit ]

*Main article: McDonnell Douglas Phantom in UK service*



An F-4J of the U.S. Navy (foreground), alongside an F-4K of the Fleet Air Arm (background) wait to be catapulted from USS *Independence*, March 1975; one of the major differences can be seen by the higher degree of the British aircraft's extendable nose wheel. Both variants were eventually used by the Royal Air Force

The United Kingdom bought versions based on the U.S. Navy's F-4J for use with the Royal Air Force and the Royal Navy's Fleet Air Arm. The UK was the only country outside the United States to operate the Phantom at sea, with them operating from HMS *Ark Royal*. The main differences were the use of the British Rolls-Royce Spey engines and of British-made avionics. The RN and RAF versions were given the designation F-4K and F-4M respectively, and entered service with the British military aircraft designations Phantom FG.1 (fighter/ground attack) and Phantom FGR.2 (fighter/ground attack/reconnaissance).[208][209]

Initially, the FGR.2 was used in the ground attack and reconnaissance role, primarily with RAF Germany, while 43 Squadron was formed in the air defense role using the FG.1s that had been intended for the Fleet Air Arm for use aboard HMS *Eagle*. The superiority of the Phantom over the English Electric Lightning in terms of both range and weapons system capability, combined with the successful introduction of the SEPECAT Jaguar, meant that, during the mid-1970s, most of the ground attack Phantoms in Germany were redeployed to the UK to replace air defense Lightning squadrons.[210] A second RAF squadron, 111 Squadron, was formed on the FG.1 in 1979 after the disbandment of 892 NAS.

In 1982, during the Falklands War, three Phantom FGR2s of No. 29 Squadron were on active Quick Reaction Alert duty on Ascension Island to protect the base from air attack.[211] After the Falklands War, 15 upgraded ex-USN F-4Js, known as the F-4J(UK) entered RAF service to compensate for one interceptor squadron redeployed to the Falklands.[148]

Around 15 RAF squadrons received various marks of Phantom, many of them based in West Germany. The first to be equipped was No. 228 Operational Conversion Unit at RAF Coningsby in August 1968. One noteworthy operator was No. 43 Squadron where Phantom FG1s remained the squadron equipment for 20 years, arriving in September 1969 and departing in July 1989. During this period the squadron was based at Leuchars.[212]

The interceptor Phantoms were replaced by the Panavia Tornado F3 from the late 1980s onwards. Originally to be used until 2003,[213] it was set back to 1992 due to restructuring of the British Armed Forces[214] and the last combat British Phantoms were retired in October 1992 when No. 74(F) Squadron was disbanded.[26][212] Phantom FG.1 *XT597* was the last British Phantom to be retired on 28 January 1994, it was used as a test jet by the Aeroplane and Armament Experimental Establishment for its whole service life.[215][216]

### Civilian use [ edit ]

In 1988, Sandia National Laboratories mounted an F-4 on a "rocket sled", then crashed it into reinforced concrete to learn about the collision of aircraft with structures such as a nuclear power plant.[217]

The Collings Foundation F-4D Phantom II, with Vietnam-era "Ritchie/DeBellevue" markings, taxis at Selfridge ANGB, May 2005

An F-4D (civilian registration NX749CF) is operated by the Massachusetts-based non-profit organization Collings Foundation as a "living history" exhibit.[26][218] Funds to maintain and operate the aircraft, which is based in Houston, Texas, are raised through donations and sponsorships from public and commercial parties.[219][220]

In the 1960s, NASA used the F-4 to photograph and film Titan II missiles after their launch from Cape Canaveral after finding the Lockheed F-104 Starfighter inadequate. Jack Petry, a retired U.S. Air Force colonel, said he put his F-4 into a Mach 1.2 dive synchronized to the launch countdown, then "walked the [rocket's] contrail". Petry's Phantom stayed with the Titan for 90 seconds, reaching 68,000 feet, then broke away as the missile continued into space.[24]

NASA's Flight Research Center acquired an F-4A on 3 December 1965. It made 55 flights flying chase on X-15 missions and lifting body flights. The F-4 also supported a biomedical monitoring program involving 1,000 flights by NASA Flight Research Center aerospace research pilots and students of the USAF Aerospace Research Pilot School flying high-performance aircraft. The pilots were instrumented to record accurate and reliable data of electrocardiogram, respiration rate, and normal acceleration. In 1967, the Phantom supported a brief military-inspired program to determine whether an airplane's sonic boom could be directed and whether it could be used as a weapon of sorts, or at least an annoyance. NASA also flew an F-4C in a spanwise blowing study from 1983 to 1985, after which it was returned.[221]

## Culture [ edit ]

### Legacy [ edit ]

*"The Fabulous Phantom II"* 5000th F-4 Phantom, McDonnell Douglas promotional film

The F-4 Phantom II is widely regarded as an iconic aircraft of post-WWII military aviation, celebrated for its versatility, performance, and distinctive design. With its bold lines and commanding presence, the Phantom earned deep respect for its performance, adaptability, and symbolic weight across generations of pilots and enthusiasts. Its long service life and widespread use in multiple roles have cemented its status as a symbol of military aviation.[222] With 5,195 Phantom II aircraft produced over more than 20 years in multiple variants, the F-4 achieved huge commercial success as the best-selling supersonic fighter in U.S. history.[223] It currently remains the latest, and final aircraft model to produce "ace-status" for a member of any branch of the United States armed forces.

### Nicknames [ edit ]

F-4F
"PHANTOM II"
F-4F
PHANTOM II
THE WORLD'S LARGEST
DISTRIBUTOR
OF MIG PARTS

An F-4F on display described as the "World's largest distributor of MiG parts", because of the high number of this type of enemy aircraft shot down

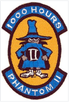
The Phantom gathered a number of nicknames during its career. Some of these names included "Snoopy", "Rhino", "Double Ugly",[224] "Old Smokey",[87] the "Flying Anvil", "Flying Footlocker", "Flying Brick", "Lead Sled", the "Big Iron Sled", and the "St. Louis Slugger" (owing to it being produced in St. Louis).[225] In recognition of its record of downing large numbers of Soviet-built MiGs,[226] it was called the "World's Leading Distributor of MiG Parts".[224] As a reflection of excellent performance in spite of its bulk, the F-4 was dubbed "the triumph of thrust over aerodynamics."[227] German *Luftwaffe* crews called their F-4s the *Eisenschwein* ("Iron Pig"), *Fliegender Ziegelstein* ("Flying Brick") and *Luftverteidigungsdiesel* ("Air Defense Diesel").[228] In the RAF it was most commonly referred to as "The Toom" (not tomb).[229] Whilst the Turkish Air Force crewmen nicknamed it as *Baba* ("Father").[230]

### Reputation [ edit ]

Imitating the spelling of the aircraft's name, McDonnell issued a series of patches. Pilots became "Phantom Phlyers", backseaters became "Phantom Pherrets", fans of the F-4 "Phantom Phanatics", and call it the "Phabulous Phantom". Ground crewmen who worked on the aircraft are known as "Phantom Phixers".[6]

Several active websites are devoted to sharing information on the F-4, and the aircraft is grudgingly admired as brutally effective by those who have flown it. Colonel (Ret.) Chuck DeBellevue reminisced, "The F-4 Phantom was the last plane that looked like it was made to kill somebody. It was a beast. It could go through a flock of birds and kick out barbeque from the back."[231] It had "A reputation of being a clumsy bruiser reliant on brute engine power and obsolete weapons technology."[232]

### The Spook [ edit ]



The Spook

The aircraft's emblem is a whimsical cartoon ghost called "The Spook", which was created by McDonnell Douglas technical artist, Anthony "Tony" Wong, for shoulder patches. The name "Spook" was coined by the crews of either the 12th Tactical Fighter Wing or the 4453rd Combat Crew Training Wing at MacDill AFB. The figure is ubiquitous, appearing on many items associated with the F-4. The Spook has followed the Phantom around the world adopting local fashions; for example, the British adaptation of the U.S. "Phantom Man"[224] is a Spook that sometimes wears a bowler hat and smokes a pipe.[233]

## Variants [ edit ]

*Main article: List of McDonnell Douglas F-4 Phantom II variants*

QF-4E AF Serial No. 74-1626 at McGuire AFB in May 2007 with an A-10 in the background

**F-4A, B, J, N and S**

Variants for the U.S. Navy and the U.S. Marine Corps. F-4B was upgraded to F-4N, and F-4J was upgraded to F-4S.

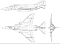
**F-110 (original USAF designation for F-4C), F-4C, D and E**

Variants for the U.S. Air Force. F-4E introduced an internal M61 Vulcan cannon. The F-4D and E were the most numerously produced, widely exported, and also extensively used under the Semi Automatic Ground Environment (SAGE) U.S. air defense system.

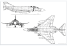
**F-4G Wild Weasel V**

A dedicated Suppression of Enemy Air Defenses (SEAD) variant for the U.S. Air Force with updated radar and avionics, converted from F-4E. The designation F-4G was applied earlier to an entirely different U.S. Navy Phantom.

**F-4 Phantom Comparisons**



F-4E Phantom (US/Foreign export)



F-4K Phantom (RN FAA/RAF only)

**F-4K and M**

Variants for the Royal Navy and Royal Air Force, respectively, built with Rolls-Royce Spey turbofan engines.

**F-4EJ and RF-4EJ**

Simplified F-4E exported to and license-built in Japan. Some modified for reconnaissance role, carrying photographic or electronic reconnaissance pods and designated RF-4EJ.

**F-4F**

Simplified F-4E exported to Germany.

**QRF-4C, QF-4B, E, G, N and S**

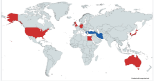
Retired aircraft converted into remote-controlled target drones used for weapons and defensive systems research by USAF and USN / USMC.

**RF-4B, C, and E**

Tactical reconnaissance variants.

## Operators [ edit ]

### Current [ edit ]



Operators: 🟦 Current 🟥 Former

**Greece**

- Hellenic Air Force – 17 F-4E AUPs (Peace Icarus 2000 variant) in service as of 2024[234][235][236]
  - Andravida Air Base, Elis
    - 338th Fighter-Bomber Squadron

**Iran**

- Islamic Republic of Iran Air Force – 60 F-4D, F-4E, and RF-4Es in service as of 2021[234][237]
  - Bandar Abbas Air Base, Hormozgan Province
    - 91st Tactical Fighter Squadron (F-4E)
  - Bushehr Air Base, Bushehr Province
    - 61st Tactical Fighter Squadron (F-4E)
  - Chabahar Konarak Air Base, Sistan and Baluchestan Province
    - 101st Tactical Fighter Squadron (F-4D)
  - Hamadan Air Base, Hamadan Province
    - 31st Tactical Reconnaissance Squadron (RF-4E)
    - 31st Tactical Fighter Squadron (F-4E)

Hellenic Air Force RF-4E Phantom II in a special color scheme, lands at RIAT 2008, UK

**Turkey**

- Turkish Air Force – 30 F-4E 2020 Terminators in service as of 2024[238]
  - Eskişehir Air Base, Eskişehir Province
    - 111 Filo

A Turkish F-4E 2020 in '60 Years' livery

### Former operators [ edit ]

**Australia**

- Royal Australian Air Force (24 leased F-4E aircraft in service between 1970 and 1973 as a stop-gap capability in the transition from Canberra bombers to the F-111C)[239]

**Egypt**

- Egyptian Air Force (F-4E 1977 to 2020)[130]

**Germany**

- German Air Force (RF-4E 1971 to 1994; F-4F 1973 to 2013; F-4E 1978 to 1992)[240]

**Greece**

- Hellenic Air Force (RF-4E 1978 to 2017)[162]

**Israel**

- Israeli Air Force (F-4E 1969 to 2004;[241] RF-4C 1970 to 1971;[242] RF-4E 1971 to 2004)[243]

**Japan**

- Japan Air Self-Defense Force (F-4EJ 1971 to 2021;[2] RF-4E 1974 to 2020; RF-4EJ 1992 to 2020)[244]

**South Korea**

- Republic of Korea Air Force (F-4D 1969 to 2010;[245] RF-4C 1989 to 2014;[246] F-4E 1977 to 2024)

**Spain**

- Spanish Air Force (F-4C 1971 to 1990; RF-4C 1978 to 2002)[247]

**Turkey**

- Turkish Air Force (RF-4E 1980 to 2015)[248]

**United Kingdom**

- Aeroplane and Armament Experimental Establishment (F-4K 1970 to 1994)[249]
- Fleet Air Arm (F-4K 1968 to 1978)[210]
- Royal Air Force (F-4M 1968 to 1992; F-4K 1969 to 1990; F-4J(UK) 1984 to 1991)[210]

**United States**

- NASA (F-4A 1965 to 1967;[250] F-4C 1983 to 1985)[221]
- United States Air Force (F-4B 1963 to 1964;[251] F-4C 1964 to 1989;[252] RF-4C 1964 to 1995;[253] F-4D 1965 to 1992; F-4E 1967 to 1991;[252] F-4G 1978 to 1996;[251] QF-4 1997 to 2016)[254][1]
- United States Marine Corps (F-4B 1962 to 1979; RF-4B 1965 to 1990; F-4J 1967 to 1984;[255] F-4N 1973 to 1985;[256] F-4S 1978 to 1992)[255]
- United States Navy (F-4A 1960 to 1968; F-4B 1961 to 1974; F-4J 1966 to 1982; F-4N 1973 to 1984; F-4S 1979 to 1987; QF-4 1983 to 2004)[257]

A Royal Australian Air Force F-4E Phantom II at RAAF Base Pearce in 1971

Egyptian Air Force F-4E Phantom IIs of the 222nd Tactical Fighter Brigade in formation with a U.S. Air Force 347th Tactical Fighter Wing F-4E Phantom II during exercise Proud Phantom

Spanish Air Force RF-4C Phantom II, 15 June 1993

### Privately owned [ edit ]

- Platinum Fighters Sale – 1 F4H-1F[258]

## Notable accidents [ edit ]

*See also: List of accidents and incidents involving military aircraft*

- On 6 June 1971, Hughes Airwest Flight 706, a McDonnell Douglas DC-9-31 collided in mid-air with a United States Marine Corps F-4B Phantom above the San Gabriel Mountains, while en route from Los Angeles International Airport to Salt Lake City. All 49 on board the DC-9 were killed, while the pilot of the F-4B was unable to eject and died when the aircraft crashed shortly afterwards. The F-4B's Radar Intercept Officer successfully ejected from the plane and parachuted to safety, being the sole survivor of the incident.
- On 9 August 1974, a Royal Air Force Phantom FGR2 was involved in a fatal collision with a civilian PA-25-235 Pawnee crop-sprayer over Norfolk, England.[259]
- On 1 October 1976, an RF-4C Phantom II impacted terrain on Holston Mountain near the Holston Mountain VORTAC, just north of Elizabethton, TN. Both crew members were killed.[260]
- On 17 August 1978, during a training exercise over the Mediterranean Sea, a US Navy F-4 shot down another US Navy F-4 with an AIM-9H. The pilots of the downed aircraft ejected and were recovered.[261]
- On 21 March 1987, Captain Dean Paul Martin, a pilot in the 163d Tactical Fighter Group of the California Air National Guard and son of entertainer Dean Martin, crashed his F-4C into San Gorgonio Mountain, California, shortly after departure from March Air Force Base. Both Martin and his weapon systems officer (WSO) were killed.[262]
- On 22 September 1987, a US Air Force RF-4C Phantom II was shot down by a US Navy F-14 Tomcat during training NATO Exercise Display Determination 87 over the Mediterranean. The RF-4C was conducting a simulated attack on USS *Saratoga* when the F-14 pilot became confused and launched a live AIM-9 Sidewinder. The RF-4C crew ejected and were recovered.[263][264]
- On 30 January 2023, a Greek Air Force F-4E Phantom II crashed into the Ionian Sea. The aircraft was conducting a training exercise when it crashed 46 km south of the Andravida Air Base. The pilot and co-pilot were killed in the crash.[265][266]

## Aircraft on display [ edit ]

*Main article: List of displayed McDonnell Douglas F-4 Phantom IIs*

As a result of its extensive number of operators and large number of aircraft produced, a significant number of F-4 Phantom IIs of a variety of variants have been preserved in museums and other locations around the world.

## Gallery [ edit ]

Structural view of partially disassembled German F-4 Phantoms (2009).

VF-96 F-4J "Showtime 100" armed with Sidewinder and Sparrow missiles (1972).

A Republic of Korea Air Force F-4D Phantom II armed with AIM-9 missiles at Daegu Air Base, 1979.

### Video resources [ edit ]

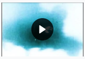
*"The Record Breaking Phantom II"* Mcdonnell aircraft promotional Film.

*"The F-4 Phantom Joins the Fleet"* (1962).

F-4J Phantom II AWG-10 guided missile trials at Pt. Mugu, California.

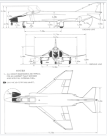
*"Ambassadors in Blue"* (1971) F-4E Phantom USAF Thunderbirds promotional film.

USAF F-4 Phantom II conventional weapons delivery dive-bombing training film.

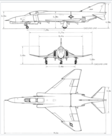


USAF tactical weapons in South-East Asia.

## Specifications (F-4E) [ edit ]



3-view line drawing of the McDonnell F-4C Phantom II



3-view line drawing of the McDonnell Douglas F-4E Phantom II

*Data from The Great Book of Fighters,*[148] Quest for Performance,[25] *Encyclopedia of USAF Aircraft,*[67] and *McDonnell F-4 Phantom: Spirit in the Skies*[267]

### General characteristics

- **Crew:** 2
- **Length:** 63 ft 0 in (19.2 m)
- **Wingspan:** 38 ft 5 in (11.7 m)
- **Width:** 27 ft 7 in (8.4[268] m) wing folded
- **Height:** 16 ft 5 in (5 m)
- **Wing area:** 530 sq ft (49.2 $m^2$)
- **Aspect ratio:** 2.77
- **Airfoil:** NACA 0006.4–64 root, NACA 0003-64 tip
- **Empty weight:** 30,328 lb (13,757 kg)
- **Gross weight:** 41,500 lb (18,824 kg)
- **Max takeoff weight:** 61,795 lb (28,030 kg)
- **Maximum landing weight:** 36,831 lb (16,706 kg)
- **Fuel capacity:** 1,994 US gal (1,660 imp gal; 7,550 L) or 12,961 lb (5,879 kg) internal, 3,335 US gal (2,777 imp gal; 12,620 L) or 21,678 lb (9,833 kg) with 2x 370 US gal (310 imp gal; 1,400 L) external tanks on the outer wing hardpoints and either a 600 or 610 US gal (500 or 510 imp gal; 2,300 or 2,300 L) tank for the center-line station.
- **Powerplant:** 2 × General Electric J79-GE-17A after-burning turbojet engines, 11,905 lbf (52.96 kN) thrust each dry, 17,845 lbf (79.38 kN) with afterburner

### Performance

- **Maximum speed:** 1,280 kn (1,470 mph, 2,370 km/h) at 40,000 ft (12,000 m)
- **Maximum speed:** Mach 2.23
- **Cruise speed:** 510 kn (580 mph, 940 km/h)
- **Combat range:** 370 nmi (420 mi, 680 km)
- **Ferry range:** 1,457 nmi (1,677 mi, 2,699 km)
- **Service ceiling:** 60,000 ft (18,000 m)
- **Rate of climb:** 41,300 ft/min (210 m/s)
- **Lift-to-drag:** 8.58
- **Wing loading:** 78 lb/sq ft (380 kg/$m^2$)
- **Thrust/weight:** 0.86 at loaded weight, 0.58 at MTOW
- **Takeoff roll:** 4,490 ft (1,370 m) at 53,814 lb (24,410 kg)
- **Landing roll:** 3,680 ft (1,120 m) at 36,831 lb (16,706 kg)

### Armament

- E-model has a 20 mm (0.787 in) M61A1 Vulcan cannon mounted internally under the nose, 640 rounds
- Up to 18,650 lb (8,480 kg) of weapons on nine external hardpoints, including general-purpose bombs, cluster bombs, TV- and laser-guided bombs, rocket pods, air-to-ground missiles, anti-ship missiles, gun pods, and nuclear weapons. Reconnaissance, targeting, electronic countermeasures and baggage pods, and external fuel tanks may also be carried.
- 4× AIM-9 Sidewinders on wing pylons, Israeli F-4 Kurnass 2000 carried Python-3, Japanese F-4EJ *Kai* carry AAM-3.
- 4× AIM-7 Sparrow in fuselage recesses, upgraded Hellenic F-4E and German F-4F ICE carry AIM-120 AMRAAM, UK Phantoms carried Skyflash missiles[269]
- 6× AGM-65 Maverick
- 4× AGM-62 Walleye
- 4× AGM-45 Shrike, AGM-88 HARM, AGM-78 Standard ARM
- 4× GBU-15, GBU-24
- 18× Mk.82, GBU-12
- 5× Mk.84, GBU-10
- 18× CBU-87, CBU-89, CBU-58
- 12x MK-20 Rockeye
- Nuclear weapons, including the B28EX, B61, B43 and B57

### Avionics

- AN/APG-30 – fire-control radar
- AN/APG-76 – Multimode synthetic-aperture radar
- AN/APQ-117 – Terrain-following radar
- AN/ASQ-153 – Part of *Pave Spike* laser designator targeting pod
- AN/ASX-1 – Target Identification System Electro-Optical (TISEO)
- AN/AVQ-23 – Part of *Pave Spike* laser designator targeting pod

## See also [ edit ]

- F-4 Phantom II in fiction
- Operation Bolo

Aviation portal

**Related development**

- McDonnell F3H Demon
- McDonnell Douglas Phantom FG.1/FGR.2/F.3

**Aircraft of comparable role, configuration, and era**

- Dassault Mirage III
- English Electric Lightning
- Mikoyan-Gurevich MiG-21
- Mikoyan-Gurevich MiG-23
- Republic F-105 Thunderchief
- Saab 35 Draken
- Shenyang J-8
- Sukhoi Su-15
- Vought XF8U-3 Crusader III

**Related lists**

- List of fighter aircraft
- List of military aircraft of the United States
- List of military electronics of the United States

## References [ edit ]

### Notes [ edit ]

1. ^ The aircraft was originally designated the AH, and later re-designated F4H, by the United States Navy, while the U.S. Air Force's original designation was "F-110A Spectre". The F-4 designation came about in the 1962 United States Tri-Service aircraft designation system for all branches of the U.S. military, unified by the order of U.S. Defense Secretary Robert McNamara. Within McDonnell Aircraft, the F-4 was referred to as Model 98.[4]

### Citations [ edit ]

1. ^ a b Devine, Jacqueline. "USAF bids farewell to F-4 Phantom" ↗.

151. ^ Klein, Andreas (2004). *Luftwaffe PHANTOMs Part 2 - The MDD*

*Military 1*. Archived from the original on 7 February 2017. Retrieved 6 February 2017.

2. ^ a b c Cenciotti, David (17 March 2021). "Japan's Last Three F-4EJ Phantoms Have Just Flown For The Last Time". *The Aviationist*. Archived from the original on 18 March 2021. Retrieved 17 March 2021.
3. ^ a b Kim, Hyun-Bin (7 June 2024). "South Korea bids farewell to legendary F-4 Phantom fighter jet". *The Korea Times*. Retrieved 7 June 2024.
4. ^ a b c Swanborough and Bowers 1976, p. 301.
5. ^ "Historical Snapshot: F-4 Phantom II Fighter". Boeing. Archived from the original on 14 August 2020. Retrieved 29 August 2020.
6. ^ a b c d e f "F-4 Phantoms Phabulous 40th". Boeing. Archived from the original on 29 June 2011. Retrieved 27 November 2012.
7. ^ "First to Last". *Phabulous 40th*. Boeing. Archived from the original on 13 December 2009. Retrieved 19 November 2007.
8. ^ Wragg, David W. (1973). *A Dictionary of Aviation* (first ed.). Osprey. p. 122. ISBN 9780850451634.
9. ^ a b "Phantom 'Phirsts' ". *Phabulous 40th*. Boeing. Archived from the original on 29 June 2011. Retrieved 27 November 2012..
10. ^ a b c d e f g h "World Record Holder". *Phabulous 40th*. Boeing. Archived from the original on 6 July 2009. Retrieved 14 December 2007.
11. ^ a b c Dorr and Bishop 1996, pp. 200–201.
12. ^ a b Dorr and Bishop 1996, pp. 188–189.
13. ^ Donald Spring 1991, p. 26.
14. ^ Donald Summer 1991, p. 22.
15. ^ a b Lake 1992, p. 190.
16. ^ Lake 1992, p. 203.
17. ^ "Equipment". Hellenic Air Force. Archived from the original on 20 March 2025. Retrieved 28 June 2025.
18. ^ Iddon, Paul. "F-4 Drop Tank Incident Underscores Turkey's Need To Replace Its Aging Phantom Fleet". *Forbes*. Retrieved 10 July 2025.
19. ^ Thornborough and Davies 1994, p. 13.
20. ^ Thornborough and Davies 1994, p. 11.
21. ^ Dorr 2008, p. 61.
22. ^ "Phabulous 40th: Phantom Development." Archived 7 September 2009 at the Wayback Machine *1978 Commemorative Book*. Boeing. Retrieved: 14 February 2008.
23. ^ Lake 1992, p. 15.
24. ^ a b c d e f g h i j Joiner, Stephen (March 2015). "What Couldn't the F-4 Phantom Do?". *Air & Space*. Archived from the original on 6 June 2020. Retrieved 5 June 2020.
25. ^ a b c Loftin, Laurence K. *Quest for Performance: The Evolution of Modern Aircraft* SP-468. Archived 13 June 2006 at the Wayback Machine Washington, DC: *National Aeronautics and Space Administration, History Office, Scientific and Technical Information Branch*, 1985. Retrieved: 19 November 2007.
26. ^ a b c d e f g h i j k Donald and Lake 2002, pp. 6, 8.
27. ^ Angelucci 1987, p. 316.
28. ^ Lake 1992, p. 21.
29. ^ a b c Eden 2004, p. 278.
30. ^ Knaack 1978, p. 266.
31. ^ Gunston 1979, p.246.
32. ^ "AWG-10 Radar, Antenna Control, Type C-8778/APG-59". Archived from the original on 14 July 2014. Retrieved 10 June 2014.
33. ^ "McDonnell F-4J Phantom II". Archived from the original on 14 July 2014. Retrieved 10 June 2014.
34. ^ "Project Bee Line". Archived from the original on 3 May 2012.
35. ^ Gunston 1981, p. 233.
36. ^ Dorr 1987, p. 39.
37. ^ "Directory: World Air Forces". Archived 11 October 2017 at the Wayback Machine *Flight International*, 11–17 November 2008, pp. 52–76.
38. ^ "Final flight farewells Phantom". *Holloman Air Force Base*. 21 December 2016. Archived from the original on 24 April 2017. Retrieved 23 April 2017.
39. ^ Iddon, Paul. "The Death Spiral Of Iran's Unique And Antique Air Force". *Forbes*. Retrieved 26 June 2026.
40. ^ "This day in aviation: 6 December 1959" Archived 26 April 2014 at the Wayback Machine *Bryan R. Swopes*. Retrieved: 25 April 2014.
41. ^ Lake 1992, pp. 16, 17.
42. ^ “Naval Aviation News”, July 1961, p. 6
43. ^ Stein, Alan J. "Seattle native Dick Gordon orbits the moon on 18 November 1969." Archived 30 May 2008 at the Wayback Machine *HistoryLink.org*, 13 June 1999. Retrieved: 13 February 2008.
44. ^ Grossnick, Roy A. "Part 9 – The Sixth Decade 1960–1969." Archived 27 February 2008 at the Wayback Machine *history.navy.mil*. Retrieved: 21 July 2010.
45. ^ "United States Naval Aviation 1910–1995." Archived 12 September 2012 at the Wayback Machine Naval Historical Center, 1997. Retrieved: 21 July 2010.
46. ^ "McDonnell F-4A (F4H-1) Phantom II 'Sageburner'." *National Air and Space Museum,* Smithsonian Institution. Retrieved: 22 August 2011.
47. ^ McComb, *Enterprise Journal*, USM.
48. ^ "22 November 1961". *This Day in Aviation*. 22 November 2016. Archived from the original on 15 March 2017. Retrieved 15 March 2017.
49. ^ "22 November 1961". 22 November 2021. Archived from the original on 15 March 2017. Retrieved 15 March 2017.
50. ^ Thornborough and Davies 1994, p. 15.
51. ^ "Naval Aviation News", April 1962, p. 12
52. ^ "Naval Aviation News", June 1962, p. 12
53. ^ "USN Photo 1143454". Archived from the original on 7 May 2022. Retrieved 7 May 2022.
54. ^ "US Navy and US Marine Corps BuNos--Third Series (145062 to 150138)". *www.joebaugher.com*. Archived from the original on 17 June 2010. Retrieved 7 May 2022.
55. ^ “Naval Aviation News”, April 1962, p. 12
56. ^ ”Naval Aviation News”, June 1962, p. 12
57. ^ "Phabulous 40th: Current Uses of Titanium: F-4." Archived

*F-4F Phantom II in German Air Force Service 1982-2003*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 13, 14. ISBN 3-935687-07-9.

152. ^ Klein, Andreas (2004). *Luftwaffe PHANTOMs Part 2 - The MDD F-4F Phantom II in German Air Force Service 1982-2003*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 14–20. ISBN 3-935687-07-9.
153. ^ List 2006, p. 51.
154. ^ Jan de Ridder, Dirk. "German Phantoms still going strong." *AirForces Monthly magazine,* June 2008 issue, p. 40.
155. ^ "German air force to bid 'Pharewell' to last F-4Fs" Archived 30 June 2013 at the Wayback Machine. Flightglobal.com, 26 June 2013.
156. ^ "Auf Wiedersehn, Phantom!" Archived 2 January 2014 at the Wayback Machine Aviation Week blog, 1 July 2013.
157. ^ Polderman, Robin (21 December 2024). "Greek Ghosts Turn 50". *The Unit Report*. Retrieved 13 June 2025.
158. ^ "McDonnell Douglas F-4E Phantom II (x2)". *www.airtattoo.com*. Retrieved 13 June 2025.
159. ^ "How the Hellenic Air Force is still flying F-4 Phantoms". *www.key.aero*. 13 July 2022. Retrieved 13 June 2025.
160. ^ Dimitropoulos, K. *F-4 Phantom* (in Greek). Athens: Constantinidis Publications, 1997. ISBN 960-8426-01-4.
161. ^ Manakanatas, D. and D. Stergiou. *Phantom F-4* (in Greek). Athens, Epikoinonies S.A., 2002.
162. ^ a b Bozinovski, Igor (8 May 2017). "Hellenic Air Force retires recce squadron, RF-4Es". *IHS Jane's 360*. Archived from the original on 19 October 2017. Retrieved 9 May 2017.
163. ^ Lake 1992 p. 213
164. ^ "Gennadii N. Eliseev Medal record and citation". Archived from the original on 16 September 2011. Retrieved 10 February 2011.
165. ^ Fricker 2000, p. 64.
166. ^ Cooper and Bishop *Air Enthusiast* March/April 2004, pp. 5–6.
167. ^ Cooper and Bishop *Air Enthusiast* March/April 2004, pp. 7–8.
168. ^ Times, Richard Halloran and Special To the New York (6 June 1984). "2 Iranian Fighters Reported Downed by Saudi Air Force". *The New York Times*. Archived from the original on 29 January 2017. Retrieved 11 February 2017.
169. ^ a b Winer, Stuart (2 December 2014). "US official confirms Iran is bombing Islamic State". *Times of Israel*. Archived from the original on 29 December 2014. Retrieved 2 December 2014.
170. ^ "Iran jets bomb Islamic State targets in Iraq – Pentagon". *BBC News*. Archived from the original on 23 September 2018. Retrieved 20 June 2018.
171. ^ Newdick, Thomas; Banković, Živojin (19 December 2024). "Iranian F-14 Tomcat Encounter At An Airshow Unlike Any Other". *The War Zone*. Recurrent Ventures. Retrieved 18 June 2025.
172. ^ Sleiman, Nadia (2 March 2026). "Iran Update Evening Special Report, March 1, 2026". *Institute for the Study of War*.
173. ^ Lake 1992 p. 215
174. ^ Nordeen 1991, p. 99.
175. ^ "Directory: World Air Forces: Israel." Archived 7 January 2012 at the Wayback Machine *Flight International*, 16–22 November 2004. Retrieved: 14 February 2008.
176. ^ Fricker 2000, p. 85.
177. ^ Baugher, Joseph F. "McDonnell F-4EJ Phantom II." Archived 16 April 2009 at the Wayback Machine *F-4 history*. Retrieved: 31 December 2009.
178. ^ office_zzam (25 November 2020). "Japan Air Force retires its last F-4EJ Phantoms". *www.armyrecognition.com*. Retrieved 23 May 2026.
179. ^ Baugher, Joseph F, "McDonnell F-4EJ Kai Phantom II" Archived 27 August 2010 at the Wayback Machine *F-4 history*, Retrieved: 31 December 2009
180. ^ Grevatt, Jon. "Japan narrows next-generation fighter requirement choice." *Jane's Defence Industry*, 21 March 2007. Retrieved: 19 November 2007.
181. ^ "Japan likely to choose F-35 for F-X requirement – reports." Archived 8 January 2012 at the Wayback Machine *Flight International*, 14 December 2011. Retrieved: 18 December 2011.
182. ^ Kallender-Umezu, Paul Japan's Fighter Procurement Crunch 6 June 2015 Archived 22 January 2023 at the Wayback Machine *Defense News* Retrieved 15 February 2017
183. ^ Reim, Garrett (29 March 2019). "Japan's air force stood up its first operational F-35A fighter squadron, the 302nd Tactical Fighter Squadron, on 29 March". *Flight Global*. Archived from the original on 30 March 2019. Retrieved 13 July 2019.
184. ^ @jasdf_hyakuri (25 March 2020). 航空総隊司令官は、#百里基地 において、#航空総隊 直轄部隊である #偵察航空隊 の廃止に伴う #隊旗返還式 を執行しました。偵空隊の総員が整列し見守る中、偵空隊司令から総隊司令官に隊旗が返還。偵空隊は3/26をもって廃止され、1961年の創設から59年の長きに亘る栄光の歴史に幕を閉じました。 (Tweet) (in Japanese) – via Twitter.
185. ^ "Japanese RF-4E Phantoms Have Just Carried Out Their Last Flight" Archived 10 March 2020 at the Wayback Machine. theaviationist.com, 9 March 2020. Quote: "However, the 301 Squadron, also based at Hyakuri, and equipped with the grey F-4EJ “Kai” jets with the squadron emblem, a frog, on the tail, will continue to operate the Phantom for some months..."
186. ^ @jasdf_hyakuri (20 November 2020). 本日 #百里基地 は、#第３０１飛行隊 壮行行事を実施しました。48年間に及ぶ #F4 運用の終止符として、基地協力団体代表の方々、歴代飛行隊長、総隊司令官、統幕副長をはじめ多くの方々のご臨席を賜り３０１飛行隊の偉業をたたえるとともに新たな門出を祝いました。 (Tweet) (in Japanese) – via Twitter.
187. ^ "The Phinal ending of 301 Hikotai Phantom operations". *Scramble.nl*. 11 December 2020. Archived from the original on 11 December 2020. Retrieved 17 March 2021.
188. ^ Lake 1992, p. 218.
189. ^ Ye Hee Lee, Michelle (24 August 2024). "South Koreans once chipped in for fighter jets. Now, they make their own". *The Washington Post*. Retrieved 3 October 2024.
190. ^ Baugher, Joseph F. "Phantom with South Korea." Archived 23 December 2023 at the Wayback Machine *Phantom II*, April 2000. Retrieved: 27 December 2010.
191. ^ "McDonnell Douglas F-4C -Phantom II". (in Spanish) Archived 27 December 2007 at the Wayback Machine Ejército del Aire, Ministerio de Defensa, España. Retrieved: 19 November 2007.

9 September 2009 at the Wayback Machine *Boeing*, 1971. "F-4B/C 1,006 lb. 7.7% of Structure, F-J/E 1,261 lb. 8.5% of Structure". Retrieved: 14 February 2008.

58. ^ Donald and Lake 1996, p. 268.
59. ^ Dorr and Donald 1990, p. 198.
60. ^ McDonnell Douglas F-4D "Phantom II". Archived 23 September 2009 at the Wayback Machine *National Museum of the USAF*. Retrieved: 20 January 2008.
61. ^ Angelucci 1987, p. 310.
62. ^ Donald, David. *Warplanes of the Fleet*. London: AIRtime Publishing Inc., 2004. ISBN 1-880588-81-1.
63. ^ Goebel, Greg. "Phantom Over Southeast Asia." *Vectorsite.net*. Retrieved: 18 January 2008.
64. ^ a b Higham and Williams 1978.
65. ^ "McDonnell Douglas F-4S Phantom" Archived 27 February 2010 at the Wayback Machine MAPS Air Museum at Akron-Canton Airport. Retrieved: 28 March 2010.
66. ^ Richardson and Spick 1984, p. 25.
67. ^ a b c Knaack 1978
68. ^ a b Thornborough and Davies 1994, p. 260.
69. ^ Lake 1992, p. 199.
70. ^ Lake 1992, p. 200.
71. ^ Dorr 1995, p. 196.
72. ^ Burgess 1985, p. 388.
73. ^ Dorr and Bishop 1996, p. 44.
74. ^ "Navy MiG Killers" Archived 27 July 2011 at the Wayback Machine, *phantomphlyers.org*
75. ^ a b Grossnick 1997.
76. ^ Hunter, Jamie and Collens, Richard. "In Relentless Pursuit of Excellence:VX-30 Bloodhounds" (PDF). *Naval Aviation News*, September–October 2004, pp. 26–29. Retrieved: 10 November 2015.
77. ^ "Marine Fighter Attack Squadron 323".
78. ^ Eden 2004, p. 276.
79. ^ Lambert, Laura. "Operation Eagle Claw". *britannica*. Archived from the original on 18 July 2018. Retrieved 17 November 2020.
80. ^ Thigpen, Jerry L.; Hobson, James L. (2011). *The Praetorian STARShip: The Untold Story of the Combat Talon*. Air University Press. p. 210. ISBN 978-1780391977. Archived from the original on 24 January 2023. Retrieved 19 October 2020.
81. ^ Eden 2004, p. 277.
82. ^ Crowther, M.J.; Baker, Rusty. "The History of VMFA-112" (PDF). *Marine Fighter Attack Squadron 112, U.S. Marine Corps*. Archived from the original (PDF) on 27 February 2008. Retrieved 13 February 2008.
83. ^ "Fact sheet discussing the F-110." Archived 20 December 2014 at the Wayback Machine *National Museum of the U.S. Air Force*. Retrieved: 26 May 2008.
84. ^ a b "McDonnell F-110 Spectre/F-4C Phantom II". *Joe Baugher*. 12 November 2018. Retrieved 30 September 2024.
85. ^ "45 Fighter Squadron (AFRC)". *Air Force Historical Research Agency*. Retrieved 30 September 2024.
86. ^ "History of the 45th Fighter Squadron". *442d Fighter Wing*. January 2011. Retrieved 30 September 2024.
87. ^ a b Wetterhahn, Ralph (January 2009). "Where Have All the Phantoms Gone?". *Air & Space*. Archived from the original on 17 June 2013. Retrieved 9 September 2013.
88. ^ Dorr and Bishop 1996, p. 38.
89. ^ Dorr and Bishop 1996, pp. 48–49.
90. ^ Dorr and Bishop 1996, p. 232.
91. ^ Knaack 1974, p. 274.
92. ^ Swanborough and Bowers 1976, p. 188.
93. ^ Baugher, Joe. "Phantom Service with USAF." Archived 4 November 2010 at the Wayback Machine Joe Baugher's Home Page. Retrieved: 27 February 2010.
94. ^ a b c d e Correll, John T. (September 2004). "The Vietnam War Almanac" (PDF). *Air & Space Forces Magazine (with attribution to USAF Operations Report, 30 November 1973)*. Vol. 87, no. 9. Archived (PDF) from the original on 28 April 2025. Retrieved 7 February 2007.
95. ^ Thornborough and Davies 1994, p. 222.
96. ^ Hobson 2001, pp. 268–270.
97. ^ Dorr and Bishop 1996, pp. 198–199.
98. ^ "History of the Springfield Air Guard: 1984 to Present". *183d Fighter Wing*. 15 May 2006. Archived from the original on 6 May 2008. Retrieved 26 September 2024.
99. ^ "McDonnell F-4E Phantom II". *Joe Baugher*. 29 July 2020. Retrieved 26 September 2024.
100. ^ Kirk, R., & Lihani, R. (Producers). (8 February 29). Dogfights "Supersonic" [Transcript, Television series episode]. In Dogfights. Houston, Texas: The History Channel.
101. ^ Handley, Phil. FU Hero Archived 17 February 2015 at the Wayback Machine. fighterpilotuniversity.com. Retrieved
102. ^ "The 1989 coup attempt: Unsung heroism, unmasked ploys". December 2021.
103. ^ Gordon, Michael R. (2 December 1989). "U.S. Says Bush Used Fighter Jets to Save Aquino from Rebel Siege". *The New York Times*.
104. ^ "Phabulous 40th: Last to Serve." Archived 13 December 2009 at the Wayback Machine *Boeing*. Retrieved: 19 November 2007.
105. ^ Bosco, Albert (23 October 2002). "Team Target' keeps Air Force on mark" (Press release). Tyndall Air Force Base: U.S. Air Force. Archived from the original on 1 July 2015. Retrieved 29 June 2015.
106. ^ "QF-4 Aerial Target". *Air Force*. Retrieved 25 September 2024.
107. ^ "82nd Aerial Target Squadron". *f-106deltadart.com*. Retrieved 25 September 2024.
108. ^ Carrara 2006, p. 48.
109. ^ a b c d PICTURE: US Air Force gets final QF-4 aerial target Archived 14 December 2013 at the Wayback Machine – Flightglobal.com, 5 December 2013
110. ^ "USAF receives last QF-4 full-scale aerial target". *Airforce Technology*. 3 December 2013. Retrieved 25 September 2024.
111. ^ a b "End of a Legend". *key.aero*. 26 January 2017. Retrieved 26 September 2024.

192. ^ Wierenga, Eddy. "Spanish Air Arms Other Air Forces, Ejército del Aire, FAMET, Armada." Archived 30 November 2007 at the Wayback Machine *Scramble (magazine)*. Retrieved: 19 November 2007.
193. ^ a b Fricker 2000, p. 88.
194. ^ "Turkey kills almost 100 terrorists in offensives led by top commanders." Archived 23 October 2011 at the Wayback Machine *Today's Zaman*, 23 October 2011.
195. ^ "Syrian military says it downed Turkish fighter jet." Archived 3 October 2014 at the Wayback Machine *BBC*, 23 June 2012.
196. ^ "Turkey to push NATO on jet downing by Syria." *CBC*, 25 June 2012.
197. ^ "Turkish jet downed by Syria in international airspace." Archived 28 July 2018 at the Wayback Machine *BBC*. Retrieved: 24 June 2012.
198. ^ Iddon, Paul. "Where Can Turkey Buy Stopgap Fighter Jets For Its Aging Air Force?". *Forbes*. Archived from the original on 16 May 2021. Retrieved 9 July 2021.
199. ^ Tegle, Eric (28 November 2021). "Turkey's Request To Buy F-16s Reminds Us That F-4 Phantoms Aren't Ghosts Yet". *Forbes*. Archived from the original on 21 June 2022. Retrieved 21 June 2022.
200. ^ Harro Ranter. "ASN Aircraft accident 24-FEB-2015 McDonnell Douglas RF-4E Phantom II". *aviation-safety.net*. Archived from the original on 2 April 2015. Retrieved 7 April 2015.
201. ^ "Two Turkish fighter jets crash, killing four". *hurriyetdailynews.com*. 25 February 2015. Archived from the original on 2 April 2015. Retrieved 7 April 2015.
202. ^ "Four dead as two Turkish military jets crash: official". *The Daily Star Newspaper – Lebanon*. Archived from the original on 2 April 2015. Retrieved 7 April 2015.
203. ^ "Turkish F-4 Phantom crashes, two dead". *janes.com*. Archived from the original on 8 April 2015. Retrieved 7 April 2015.
204. ^ "Turkish military aircraft crashes, two pilots dead". *DailySabah*. 5 March 2015. Archived from the original on 2 April 2015. Retrieved 7 April 2015.
205. ^ "Turkey hits ISIS capital, targets amid war with PKK". *Daily Sabah*. 13 September 2015. Archived from the original on 13 September 2015. Retrieved 14 September 2015.
206. ^ "Turkish military destroys PKK targets in Northern Iraq". *Daily Sabah*. 13 January 2016. Archived from the original on 4 March 2016. Retrieved 12 March 2016.
207. ^ "Turkish military hit 44 PKK targets in northern Iraq". 15 November 2015. Archived from the original on 22 November 2015. Retrieved 12 March 2016.
208. ^ Donald 1999, p. 11.
209. ^ Donald 1999, p. 5.
210. ^ a b c "History". *McDonnell-Douglas/BAC F-4K/M Phantom II*. Thunder and Lightnings. Archived from the original on 1 June 2013. Retrieved 2 April 2013.
211. ^ Burden et al. 1986, pp. 417–419.
212. ^ a b Jefford 2001.
213. ^ PHANTOM PHINALE... The 'Phinal' year of the Phantom in RAF service
214. ^ "How the RAF Nearly Kept the Phantom". *key.aero*. 14 October 2022. Retrieved 9 August 2023.
215. ^ "Not a Lot of People Know That..." *Royal Air Force Museum*. Archived from the original on 13 July 2021. Retrieved 13 July 2021.
216. ^ "XT897". *British Phantom Aviation Group*. Archived from the original on 13 July 2021. Retrieved 13 July 2021.
217. ^ Footage of 1988 rocket-sled test Archived 21 July 2011 at the Wayback Machine, "Slow motion video." Archived 21 July 2011 at the Wayback Machine *Sandia National Laboratories* via sandia.gov. Retrieved: 2 March 2011.
218. ^ "McDonnell Douglas Phantom II." Archived 29 October 2007 at the Wayback Machine *Collings Foundation*. Retrieved: 19 November 2007.
219. ^ "Collings Foundation Background." Archived 25 October 2012 at the Wayback Machine *Collings Foundation*. Retrieved 11 January 2008.
220. ^ "Sponsors." Archived 1 September 2009 at the Wayback Machine *Collings Foundation*. Retrieved: 11 January 2008.
221. ^ a b "NASA Dryden F-4 Graphics Collection." Archived 1 November 2009 at the Wayback Machine *NASA*. Retrieved: 1 August 2009.
222. ^ Anderson, William E. (1998). *Phantom in Combat*. Schiffer Publishing. ISBN 978-0764303463.
223. ^ "How the F-4 Phantom II Became the Best-Selling Supersonic Fighter Jet Plane". *ig.space*. Retrieved 11 July 2025.
224. ^ a b c "Phabulous 40th: Gee Whiz!" Archived 15 December 2009 at the Wayback Machine *Boeing*. Retrieved: 20 January 2008.
225. ^ "Basic Aircraft Wisdom and Aircraft Nicknames." Archived 2 November 2007 at the Wayback Machine *Bluejacket.com*. Retrieved: 19 November 2007.
226. ^ Thornborough and Davies, 1994. p. 202.
227. ^ "More QF-4s – And A New Trick for Old Dogs?" Archived 24 January 2008 at the Wayback Machine *Defense Industry Daily*, 22 January 2008. Retrieved: 26 January 2008. Quote: "These large 2-seat multi-role fighters were a triumph of thrust over aerodynamics, and formed the mainstay of the USAF and U.S. Navy fleets for many years."
228. ^ "Zwei Alarmrotten mit dem 'Luftverteidigungsdiesel'" (in German) Archived 31 May 2008 at the Wayback Machine *Hamburger Abendblatt*, 25 November 2004. Retrieved: 26 January 2008.
229. ^ Aviation Classics, Issue 25, 3 October 2016, ISBN 978-1-909128-32-3
230. ^ " "Duble Çirkin" F-4 Phantom II Neden Bu Kadar Çok Sevilen Bir Uçak?". *Strasam* (in Turkish). 11 April 2023. Retrieved 20 January 2025.
231. ^ Leone, Dario (23 February 2017). "Saying Goodbye to Old Smokey: "The Phinal Phantom Phlight" ". *The Aviation Geek Club*. Archived from the original on 20 September 2021. Retrieved 14 September 2021.
232. ^ Roblin, Sebastien (25 June 2016). "The F-4 Is a Great Fighter With a Bad Reputation". *Warisboring.com*. Archived from the

112. ^ Echols, SrA Alex (28 May 2015). "Final Tyndall QF-4 takes off". 325th Fighter Wing Public Affairs. Archived from the original on 1 July 2015. Retrieved 28 June 2015.
113. ^ Phenol unmanned Phantom Mission 31 August 2016 Archived 20 November 2016 at the Wayback Machine *combataircraft.net* Retrieved 20 November 2016
114. ^ Rogoway, Tyler (31 August 2016). "The Phantastic QF-4 Phantom Full-Scale Target Drone Takes its Phinal Unmanned Phlight". *The War Zone*. Retrieved 25 September 2024.
115. ^ [1] ' Retrieved 23 December 2016
116. ^ USAF QF-4s set for 2017 out of service date Archived 24 December 2015 at the Wayback Machine – Flightglobal.com, 9 November 2015
117. ^ Melampy 2011, pp. 38–39.
118. ^ "MiG-21 against the Phantom". Archived from the original on 29 November 2014. Retrieved 14 November 2014.
119. ^ Anderton 1987, pp. 70–71.
120. ^ "Nga nói gì về cuộc đấu MiG-21 và F-4 ở Việt Nam(2)". *Kienthuc.net.vn*. 27 December 2013. Archived from the original on 21 October 2014. Retrieved 14 November 2014.
121. ^ Migs over North Vietnam: The Vietnam People's Air Force in Combat, 1965–75, Stackpole Military History
122. ^ *a b c* "Mig-21 against the Phantom". Archived from the original on 29 November 2014.
123. ^ *a b* "Vietnamese Air-to-Air Victories, Part 2". Archived from the original on 4 March 2013. Retrieved 14 November 2014.
124. ^ "System Failure Case Studies: Forrestal in Flames" (PDF). NASA. December 2007. Retrieved 4 December 2023.
125. ^ Caiella, J. M. (August 2022). "Dissecting a Carrier Disaster". *Naval History Magazine*. U.S Naval Institute. Retrieved 4 December 2023.
126. ^ *a b* Lake 1992, p. 209.
127. ^ "RAAF Aircraft Series 3 A69 Phantom". Archived 21 December 2008 at the Wayback Machine *RAAF Museum Point Cook*. Retrieved: 14 November 2008.
128. ^ Fricker 2000, p. 59.
129. ^ Fricker 2000, p. 60.
130. ^ *a b* "Phantom wing finally re-equipped with F-16s". *Arabian Aerospace*. 23 July 2020. Archived from the original on 12 April 2021. Retrieved 18 March 2021.
131. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 3–7. ISBN 3-935687-08-7.
132. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 7–10. ISBN 3-935687-08-7.
133. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. p. 10. ISBN 3-935687-08-7.
134. ^ Lake 1992, p. 210.
135. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. p. 14. ISBN 3-935687-08-7.
136. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 34–38. ISBN 3-935687-08-7.
137. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 38, 39. ISBN 3-935687-08-7.
138. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. p. 40. ISBN 3-935687-08-7.
139. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. p. 41. ISBN 3-935687-08-7.
140. ^ Fricker 2000, p. 80.
141. ^ Klein, Andreas; Jahnke, Rolf (2004). *Luftwaffe PHANTOMs Part 3 - The MDD RF-4E Phantom II in German Air Force Service*. Erlangen, Germany: AirDOC Aircraft Documentations. p. 42. ISBN 3-935687-08-7.
142. ^ Klein, Andreas (2004). *Luftwaffe PHANTOMs Part 1 - The MDD F-4F Phantom II in German Air Force Service 1973-1982*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 17, 18. ISBN 3-935687-06-0.
143. ^ Klein, Andreas (2004). *Luftwaffe PHANTOMs Part 1 - The MDD F-4F Phantom II in German Air Force Service 1973-1982*. Erlangen, Germany: AirDOC Aircraft Documentations. p. 10. ISBN 3-935687-06-0.
144. ^ Klein, Andreas (2004). *Luftwaffe PHANTOMs Part 1 - The MDD F-4F Phantom II in German Air Force Service 1973-1982*. Erlangen, Germany: AirDOC Aircraft Documentations. p. 12. ISBN 3-935687-06-0.
145. ^ Lake 1992 p. 210.
146. ^ Fricker 2000, p. 81.
147. ^ Klein, Andreas (2004). *Luftwaffe PHANTOMs Part 1 - The MDD F-4F Phantom II in German Air Force Service 1973-1982*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 10, 11. ISBN 3-935687-06-0.
148. ^ *a b c d* Green and Swanborough 2001.
149. ^ Klein, Andreas (2004). *Luftwaffe PHANTOMs Part 2 - The MDD F-4F Phantom II in German Air Force Service 1982-2003*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 2, 3. ISBN 3-935687-07-9.
150. ^ Klein, Andreas (2004). *Luftwaffe PHANTOMs Part 2 - The MDD F-4F Phantom II in German Air Force Service 1982-2003*. Erlangen, Germany: AirDOC Aircraft Documentations. pp. 4–7. ISBN 3-935687-07-9.

original on 14 February 2020. Retrieved 3 March 2020.

233. ^ "Phabulous 40th: UK Spook." Archived 6 July 2009 at the Wayback Machine *Boeing*. Retrieved: 13 February 2008.
234. ^ *a b* Taghvaee, Babak (April 2021). "Samurai Sunset". *Air Forces Monthly*. Key Publishing Ltd. pp. 48–49.
235. ^ "338 Squadron". *Hellenic Air Force*. Hellenic Air Force. Archived from the original on 10 January 2025. Retrieved 10 January 2025. "At the end of 1997, after 23 years of service, HAF decided to upgrade the aircraft purchased under the programs Peace Icarus I and Peace Icarus II. On December 18th, 2002, the delivery ceremony of the first upgraded aircraft was held in the Hellenic Aviation Industry facilities, and so began the integration of the upgraded aircraft in 117 Combat Wing. These aircraft were delivered to 338 and 339 Squadron. After the modernization, the Squadron assumed fighter bomber missions, with results comparable to those of third generation aircraft."
236. ^ "2025 WORLD AIRFORCES". *World Air Forces*: 19. 2024. Archived from the original on 5 January 2025.
237. ^ "Orbats: Iran – Air Force". *Scramble.nl*. Archived from the original on 23 March 2021. Retrieved 22 March 2021.
238. ^ *Military Balance 2024*. IISS. 2024. ISBN 978-1032780047.
239. ^ Baugher, Joseph F. (28 September 2015). "Phantom with Royal Australian Air Force". *Joe Baugher*. Archived from the original on 30 July 2021. Retrieved 18 March 2021.
240. ^ Baugher, Joseph F. (28 September 2015). "Phantom with Luftwaffe". *Joe Baugher*. Archived from the original on 15 April 2021. Retrieved 18 March 2021.
241. ^ Baugher, Joseph F. (28 September 2015). "Phantom with Israel". *Joe Baugher*. Archived from the original on 6 September 2018. Retrieved 18 March 2021.
242. ^ Hayles, John (3 March 2007). "Israel Air Force". *Aeroflight*. Archived from the original on 16 October 2021. Retrieved 18 March 2021.
243. ^ Petrinic, Emil (March 2015). "F-4X: The Fastest Phantom". *HistoryNet*. Archived from the original on 19 May 2021. Retrieved 18 March 2021.
244. ^ "Japan retires its last McDonnell Douglas RF-4E/EJ "Photo Phantoms" recon aircraft". *Asia Pacific Defense Journal*. 18 March 2020. Archived from the original on 25 September 2020. Retrieved 20 November 2020.
245. ^ Donald, David (2 July 2010). "McDonnell F-4D Phantom Bows Out of Service". *AIN Online*. Archived from the original on 20 October 2021. Retrieved 18 March 2021.
246. ^ "세계 최후의 RF-4C 정찰기, 한국서 은퇴 비행". *NewDaily* (in Korean). 28 February 2014. Archived from the original on 16 October 2021. Retrieved 18 March 2021.
247. ^ Baugher, Joseph F. (8 May 2004). "Phantom with Spain". *Joe Baugher*. Archived from the original on 30 July 2021. Retrieved 18 March 2021.
248. ^ Ozbek, Tolga (11 March 2015). "Twin crash prompts Turkey to order RF-4E retirement". *Flight Global*. Archived from the original on 10 April 2021. Retrieved 18 March 2021.
249. ^ "Survivors". *British Phantom Aviation Group*. Archived from the original on 13 July 2021. Retrieved 13 July 2021.
250. ^ "F-4A Image Gallery". *NASA*. 27 August 2013. Archived from the original on 22 January 2023. Retrieved 25 March 2021.
251. ^ *a b* Baugher, Joseph F. (28 September 2015). "Phantom Service with USAF". *Joe Baugher*. Archived from the original on 4 November 2010. Retrieved 25 March 2021.
252. ^ *a b* Baugher, Joseph F. (30 April 2000). "Phantom with Air National Guard". *Joe Baugher*. Archived from the original on 23 October 2021. Retrieved 25 March 2021.
253. ^ "Aircraft History". 152nd Airlift Wing. 2 June 2008. Archived from the original on 8 May 2021. Retrieved 25 March 2021.
254. ^ Holcomb, Paul (26 October 2016). "QF-4 mission nearly complete". *Air Force*. Retrieved 25 September 2024.
255. ^ *a b* Baugher, Joseph F. (28 September 2015). "Phantom with US Marine Corps". *Joe Baugher*. Archived from the original on 23 October 2021. Retrieved 18 March 2021.
256. ^ "McDonnell F-4N Phantom II BuNo 152291". *Naval Air Museum Barbers Point*. Archived from the original on 19 April 2021. Retrieved 21 March 2021.
257. ^ Baugher, Joseph F. (28 September 2015). "Phantom with US Navy". *Joe Baugher*. Archived from the original on 23 October 2021. Retrieved 18 March 2021.
258. ^ "F4 Phantom for Sale: $3M for the World's Only Privately Owned Airworthy F-4 Phantom". *War History Online*. 15 September 2021. Retrieved 16 December 2024.
259. ^ "Aircraft Accident Report 9/75" (PDF). Department of Trade Accidents Investigation Branch. 1975. Retrieved 7 December 2023.
260. ^ "Accident McDonnell Douglass RF-4C". Aviation Safety Network. 2021. Retrieved 10 March 2025.
261. ^ Svajda, Dave (30 October 2018). "This U.S. Navy F-4 has been shot down by a U.S. Navy F-4 (flown by a USAF exchange pilot)". *The Aviation Geek Club*. Retrieved 7 December 2023.
262. ^ "The Son Of Singer Dean Martin Killed While Flying His Military Phantom Jet". *avstop.com*. Archived from the original on 27 September 2011. Retrieved 9 April 2011.
263. ^ Leone, Dario (21 September 2023). "[Video] That time a US Navy F-14 Tomcat shot down a USAF RF-4C Phantom II during a war game". *The Aviation Geek Club*. Retrieved 7 December 2023.
264. ^ Thompson, Mark (18 April 1988). "Downing of U.S. Jet Exposes War Games' Danger". *Chicago Tribune*. Retrieved 7 December 2023.
265. ^ Kokkinidis, Tasos (30 January 2023). "One Dead, One Missing After Fighter Jet Crashes in Greece". *Greek Reporter*. Retrieved 30 January 2023.
266. ^ Kokkinidis, Tasos (1 February 2023). "Greece Confirms Death of Second Pilot of Fallen F-4 Jet". *Greek Reporter*. Retrieved 1 May 2023.
267. ^ Lake 1992, pp. 92–111.
268. ^ *McDonnell Douglas F-4E Flight Manual* (PDF) (1979 ed.). p. 1. Archived (PDF) from the original on 18 January 2023. Retrieved 18 January 2023.
269. ^ From 1978 replacing the AIM-7 Sparrow

## Bibliography [ edit ]

- Angelucci, Enzo. *The American Fighter*. Sparkford, Somerset, UK: Haynes Publishing Group, 1987. ISBN 0-85429-635-2.
- Beit-Hallahmi, Benjamin. *The Israeli Connection: Whom Israel Arms and Why*. London: I.G. Tauris, 1987. ISBN 1-85043-069-1.

- Bishop, Farzad and Tom Cooper. *Iranian F-4 Phantom II Units in Combat (Osprey Combat Aircraft #37)*. Oxford, UK: Osprey Publishing Limited, 2003. ISBN 978-1-84176-658-4.
- Bowers, Peter M. and Enzo Angellucci. *The American Fighter*. New York: Orion Books, 1987. ISBN 0-517-56588-9.
- Burden, Rodney, Michael I. Draper, Douglas A. Rough, Colin R. Smith and David L. Wilton. *Falklands: The Air War*. London: Arms and Armour Press, 1986. ISBN 0-85368-842-7.
- Burgess, Richard E. *The Naval Aviation Guide*, 4th ed. Annapolis, Maryland: Naval Institute Press, 1985. ISBN 0-87021-409-8.
- Calvert, Denis. *Le Tigri della RAF (RAF's Tigers)*(in Italian). *Aerei magazine N.5*, Parma, Italy: Delta editrice, 1991.
- Carrara, Dino. *Phantom Targets: The USAFs Last F-4 Squadron*. *Air International*, Volume 71, no. 5, November 2006. Stamford, Lincolnshire, UK: Key Publishing, pp. 42–48.ISSN 0306-5634.
- Cooper, Tom and Farzad Bishop. *Target Saddam's Reactor: Israeli and Iranian Operations Against Iraqi Planes to Develop Nuclear Weapons*. *Air Enthusiast*, No. 110, March/April 2004. pp. 2–12.ISSN 0143-5450.
- Davies, Peter E. *USAF F-4 Phantom II MiG Killers 1965-68 (Osprey Combat Aircraft #45)*. Oxford, UK: Osprey Publishing Limited, 2004. ISBN 978-1-84176-656-0.
- Davies, Peter E. *USAF F-4 Phantom II MiG Killers 1972-73 (Osprey Combat Aircraft #55)*. Oxford, UK: Osprey Publishing Limited, 2005. ISBN 978-1-84176-657-7.
- Deurenberg, Rudd. *Shedding Light on Iranian Phantoms*. *Air Enthusiast*, No. 111, May/June 2004, p. 72. ISSN 0143-5450
- Donald, David. *RAF Phantoms*. *Wings of Fame*. London: Aerospace. Volume 15, 1999. pp. 4–21. ISBN 1-86184-033-0.
- Donald, David and Jon Lake, eds. *Desert Storm: The First Phase*. *World Air Power Journal*. London: Aerospace, Volume 5, Spring 1991.ISSN 0959-7050.
- Donald, David and Jon Lake, eds. *Desert Storm: Gulf Victory*. *World Air Power Journal*. London: Aerospace, Volume 6, Summer 1991.ISSN 0959-7050.
- Donald, David and Jon Lake, eds. *Encyclopedia of World Military Aircraft*. London: AIRtime Publishing, 1996. ISBN 1-880588-24-2.
- Donald, David and Jon Lake, eds. *McDonnell F-4 Phantom: Spirit in the Skies*. London: AIRtime Publishing, 2002. ISBN 1-880588-31-5.
- Dorr, Robert F. *Navy Phantoms in Vietnam*. *Wings of Fame*, Volume 1, 1995. London: Aerospace Publishing. ISBN 1-874023-68-9.
- Dorr, Robert F. "McDonnell F3H Demon". *Aeroplane*. Volume 36, No. 3, March 2008, pp. 58–61. London: IBC.
- Dorr, Robert F. and Chris Bishop, eds. *Vietnam Air War Debrief*. London: Aerospace Publishing, 1996. ISBN 1-874023-78-6.
- Dorr, Robert F. and Jon Lake. *Fighters of the United States Air Force*. London: Temple Press, 1990. ISBN 0-600-55094-X.
- Dorr, Robert F. *Phantoms Forever*. London: Osprey Publishing Limited, 1987. ISBN 0-85045-742-4.
- Eden, Paul ed. *The Encyclopedia of Modern Military Aircraft*. London: Amber Books Ltd, 2004. ISBN 1-904687-84-9.
- Elward, Brad and Peter Davies. *US Navy F-4 Phantom II MiG Killers 1965-70 (Osprey Combat Aircraft #26)*. Oxford, UK: Osprey Publishing Limited, 2001. ISBN 978-1-84176-163-3.
- Elward, Brad and Peter Davies. *US Navy F-4 Phantom II MiG Killers 1972-73 (Osprey Combat Aircraft #30)*. Oxford, UK: Osprey Publishing Limited, 2002. ISBN 978-1-84176-264-7.
- Freeman, CJ and Gunston, Bill Consulting ed. *The Encyclopedia of World Airpower*. Crown Publishers, 1979. ISBN 0-517-53754-0.
- Fricker, John. "Boeing /McDonnell Douglas F-4 Phantom II Current Operators". *World Air Power Journal*. London: Aerospace, Volume 40, Spring 2000. ISBN 1-86184-043-8.
- Green, William and Gordon Swanborough. *The Great Book of Fighters*. St. Paul, Minnesota: MBI Publishing, 2001. ISBN 0-7603-1194-3.
- Gimmi, Russell M. *Airman: The Life of Richard F. B. Gimmi*. Bloomington, Indiana: iUniverse, 2009. ISBN 978-1-4401-5311-2.
- Goodrum, Alastair (January–February 2004). "Down Range: Losses over the Wash in the 1960s and 1970s". *Air Enthusiast* (109): 12–17. ISSN 0143-5450.
- Grossnick, Roy and William J. Armstrong. *United States Naval Aviation, 1910–1995*. Annapolis, Maryland: Naval Historical Center, 1997. ISBN 0-16-049124-X.
- Gunston, Bill ed. *The Illustrated History of Fighters*. New York, New York: Exeter Books Div. of Simon Schuster, 1981. ISBN 0-89673-103-0.
- Gunston, Bill Consulting ed. *The Encyclopedia of World Airpower*. Crown Publishers, 1979. ISBN 0-517-53754-0.
- Higham, Robin and Carol Williams. *Flying Combat Aircraft of USAAF-USAF (Vol.2)*. Manhattan, Kansas: Sunflower University Press, 1978. ISBN 0-8138-0375-6.
- Hobson, Chris. *Vietnam Air Losses, USAF, USN, USMC, Fixed-Wing Aircraft Losses in Southeast Asia 1961–1973*. North Branch, Minnesota: Specialty Press, 2001. ISBN 1-85780-115-6.
- Howarth, Alan. *Spanish Phantoms and Their Legacy*. Air Enthusiast 115, January–February 2005, p. 74ISSN 0143-5450
- Jefford, C.G. *RAF Squadrons: A Comprehensive Record of the Movement and Equipment of All RAF Squadrons and Their Antecedents Since 1912:*. Shrewsbury, UK: Airlife Publishing, 2nd edition, 2001. ISBN 1-84037-141-2
- Jones, Lloyd S. *U.S. Fighters: 1925–1980s*. Fallbrook, California: Aero Publishers, Inc., 1975. ISBN 0-8168-9200-8.
- Knaack, Marcelle Size. *Encyclopedia of U.S. Air Force Aircraft and Missile Systems: Volume 1 Post-World War II Fighters 1945–1973*. Washington, DC: Office of Air Force History, 1978. ISBN 0-912799-59-5.
- Lake Jon. *McDonnell F-4 Phantom: Spirit in the Skies*. London: Aerospace Publishing, 1992. ISBN 1-880588-04-8.
- List, Friedrich. "German Air Arms Review". *Air International*, Volume 70, No. 5, May 2006, pp. 50–57. Stamford, Lincolnshire, UK: Key Publishing.ISSN 0306-5634.
- Melampy, Jake. "Phantoms West". *Air International*, Volume 80, No. 1, January 2011, pp. 36–38. Stamford, Lincolnshire, UK: Key Publishing.ISSN 0306-5634.
- Nordeen, Lon. *Fighters Over Israel: The Story of the Israeli Air Force from the War of Independence to the Bekaa Valley*. London: Guild Publishing, 1991. ISBN 1-85367-098-7.
- Richardson, Doug and Mike Spick. *F-4 Phantom II (Modern Fighting Aircraft, Volume 4)* . New York: Arco Publishing, 1984. ISBN 0-668-06068-9.
- Swanborough, Gordon and Peter Bowers. *United States Military Aircraft Since 1909*. Washington, District of Columbia: Smithsonian, 1989. ISBN 0-87474-880-1.
- Swanborough, Gordon and Peter Bowers. *United States Navy Aircraft since 1911*. London: Putnam, 1976. ISBN 0-370-10054-9.
- Taylor, Michael J.H. *Jane's American Fighting Aircraft of the 20th century*. New York: Mallard Press, 1991. ISBN 0-7924-5627-0.
- Thetford, Owen. *British Naval Aircraft since 1912*. London: Putnam, Fourth Edition, 1994, pp. 254–255. ISBN 0-85177-861-5.
- Thornborough, Anthony M. and Peter E. Davies. *The Phantom Story*. London: Arms and Armour Press, 1994. ISBN 1-85409-121-2.
- Wagner, Ray. *American Combat Planes, Third Enlarged Edition*. New York: Doubleday, 1982. ISBN 0-385-13120-8.
- Wilson, Stewart. *Phantom, Hornet and Skyhawk in Australian Service*. Weston Creek, ACT, Australia: Aerospace Publications, 1993. ISBN 1-875671-03-X.

## External links [ edit ]

Wikimedia Commons has media related to: ***F-4 Phantom II*** (category)

Wikisource has several original texts related to ***Audio recordings and transcripts with comments of actual Wild Weasel combat missions over Vietnam.***.

- F-4 Phantom II history page on Boeing.com
- F-4 Phantom II Society site
- PhantomF4K.org – Fleet Air Arm – Royal Navy site
- F-4.nl site
- Countering Israeli Reaction to F-4 Sales to Saudi Arabia and Kuwait Archived 20 March 2012 at the Wayback Machine
- 8th Tactical Fighter Wing site
- F-4 Phantom II articles and publications, theaviationindex.com
- The Phantom page with images on fas.org
- ""Silver Lobos' fly into retirement", af.mil". *Air Force*. 22 December 2004. Archived from the original on 12 February 2020. Retrieved 23 December 2004.
- "The Phantom Turns 50" article at Fence Check site Archived 19 December 2008 at the Wayback Machine
- F-4 Phantom page on Aerospaceweb.org
- RAF Phantom Losses
- The Phantom Zone Archived 3 July 2022 at the Wayback Machine
- A film clip of McDonnell Douglas F-4 Phantom II is available for viewing at the Internet Archive
- Phantom 50th Anniversary Slideshow

| V · T · E | **McDonnell and McDonnell Douglas aircraft and spacecraft** | [show] |
|---|---|---|
| | **Designation sequences for this aircraft:** | [show] |
| | **Authority control databases** | [show] |

Categories: McDonnell Douglas F-4 Phantom II | McDonnell aircraft | 1950s United States fighter aircraft | Aircraft first flown in 1958 | Carrier-based aircraft | Low-wing aircraft | Twinjets | Third-generation jet fighters | Aircraft with retractable tricycle landing gear

This page was last edited on 11 July 2026, at 14:53 (UTC). Page was rendered with Parsoid.

Text is available under the Creative Commons Attribution-ShareAlike 4.0 License; additional terms may apply. By using this site, you agree to the Terms of Use and Privacy Policy. Wikipedia® is a registered trademark of the Wikimedia Foundation, Inc., a non-profit organization.